

BFS Traversal

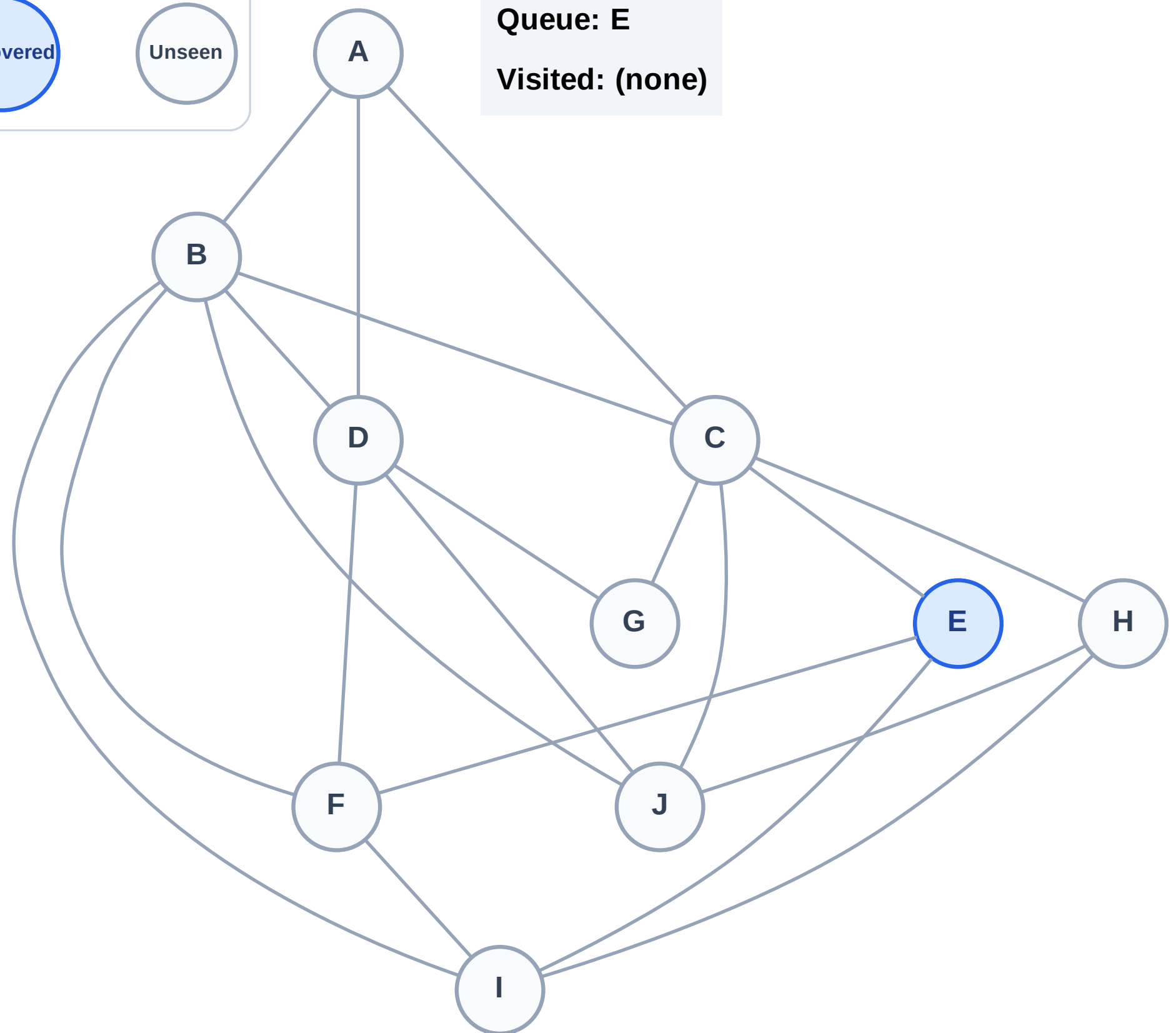
Step 1: Start at E

Legend



Queue: E

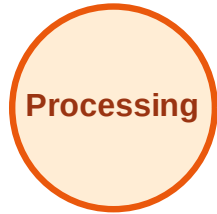
Visited: (none)



BFS Traversal

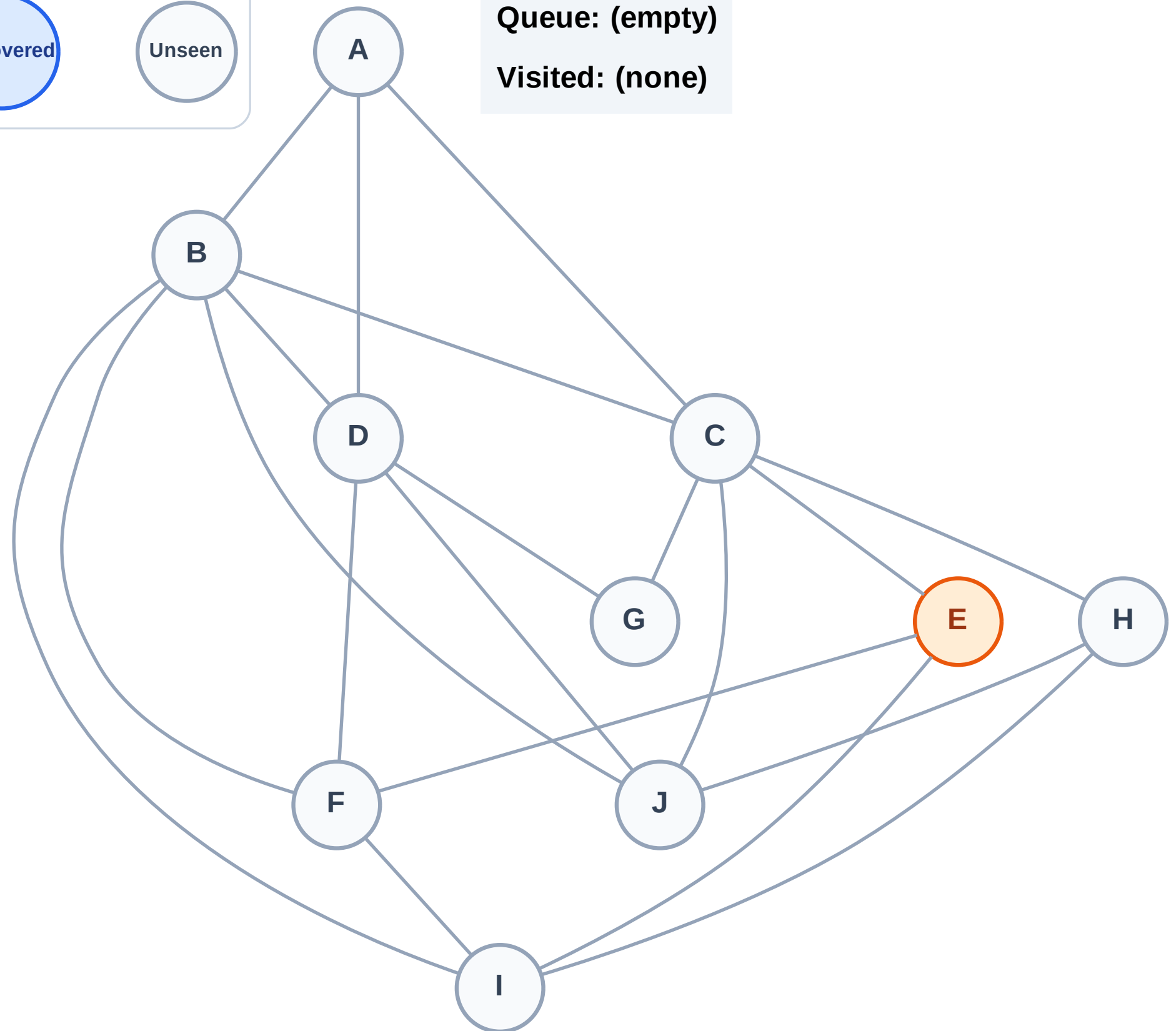
Step 2: Process node E

Legend



Queue: (empty)

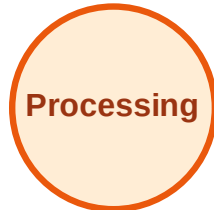
Visited: (none)



BFS Traversal

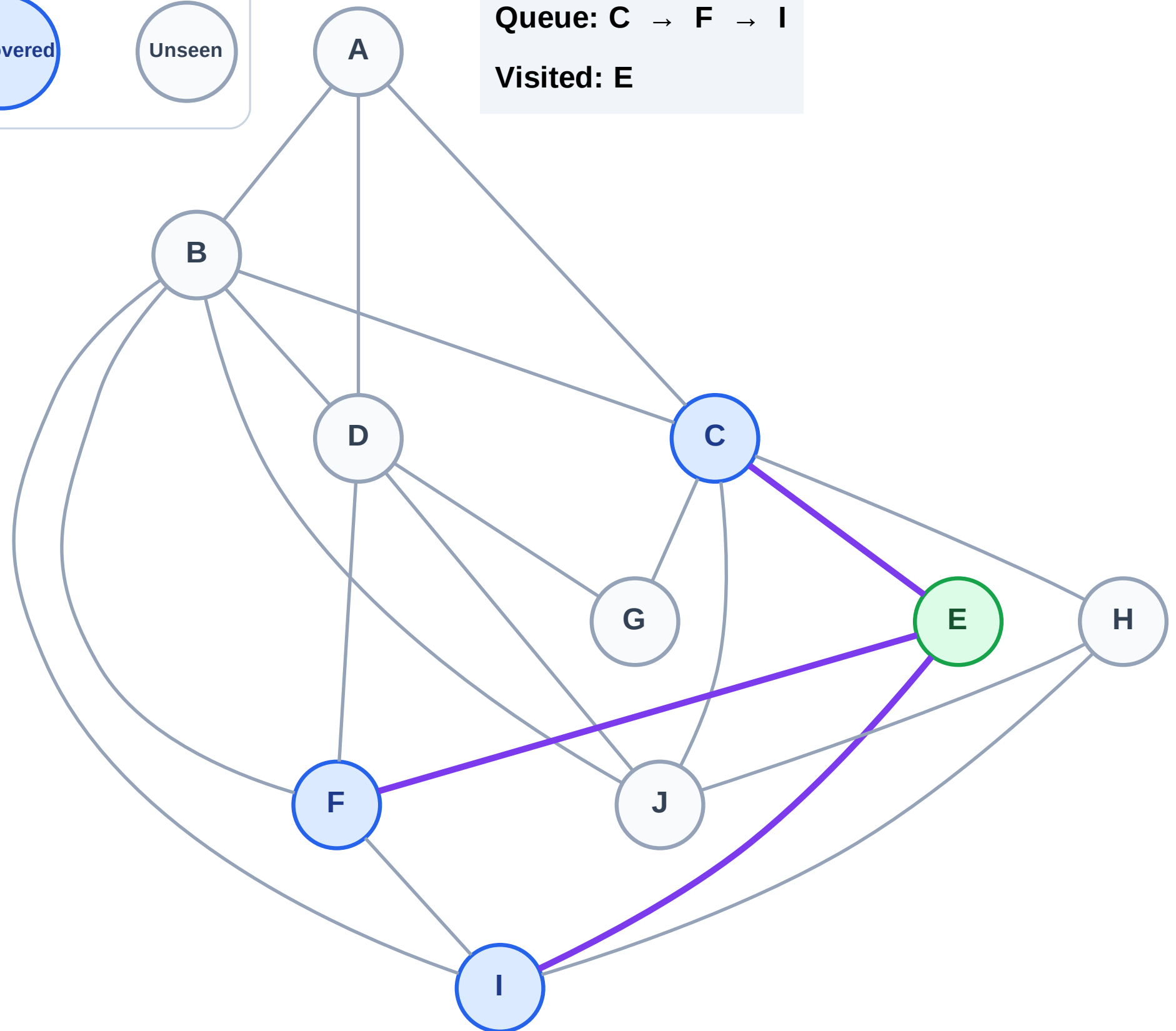
Step 3: Discover C, F, I from E

Legend



Queue: C → F → I

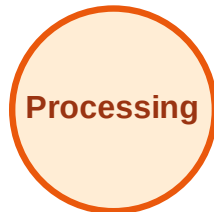
Visited: E



BFS Traversal

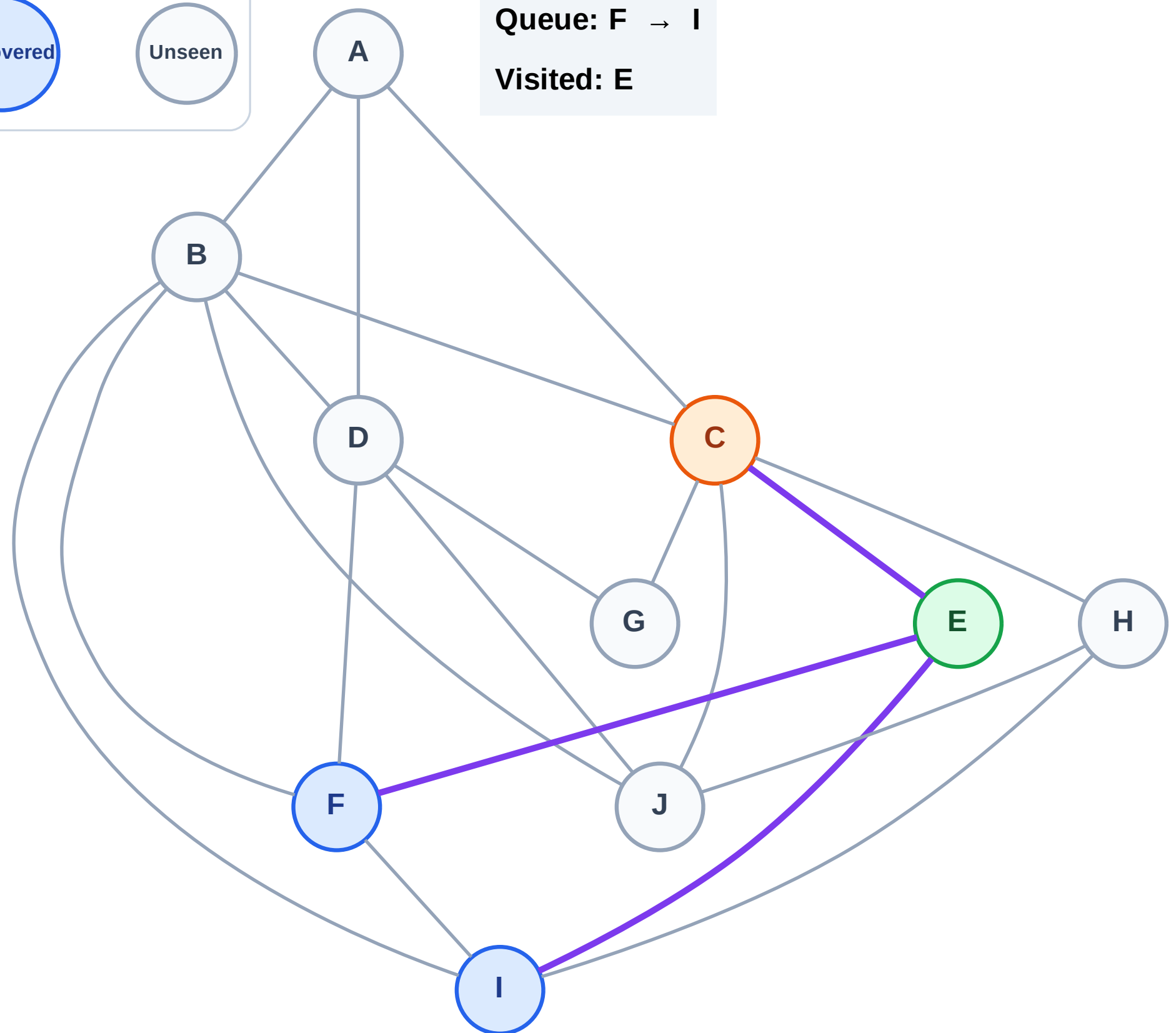
Step 4: Process node C

Legend



Queue: F → I

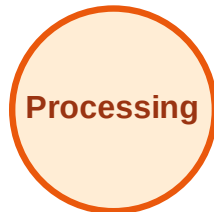
Visited: E



BFS Traversal

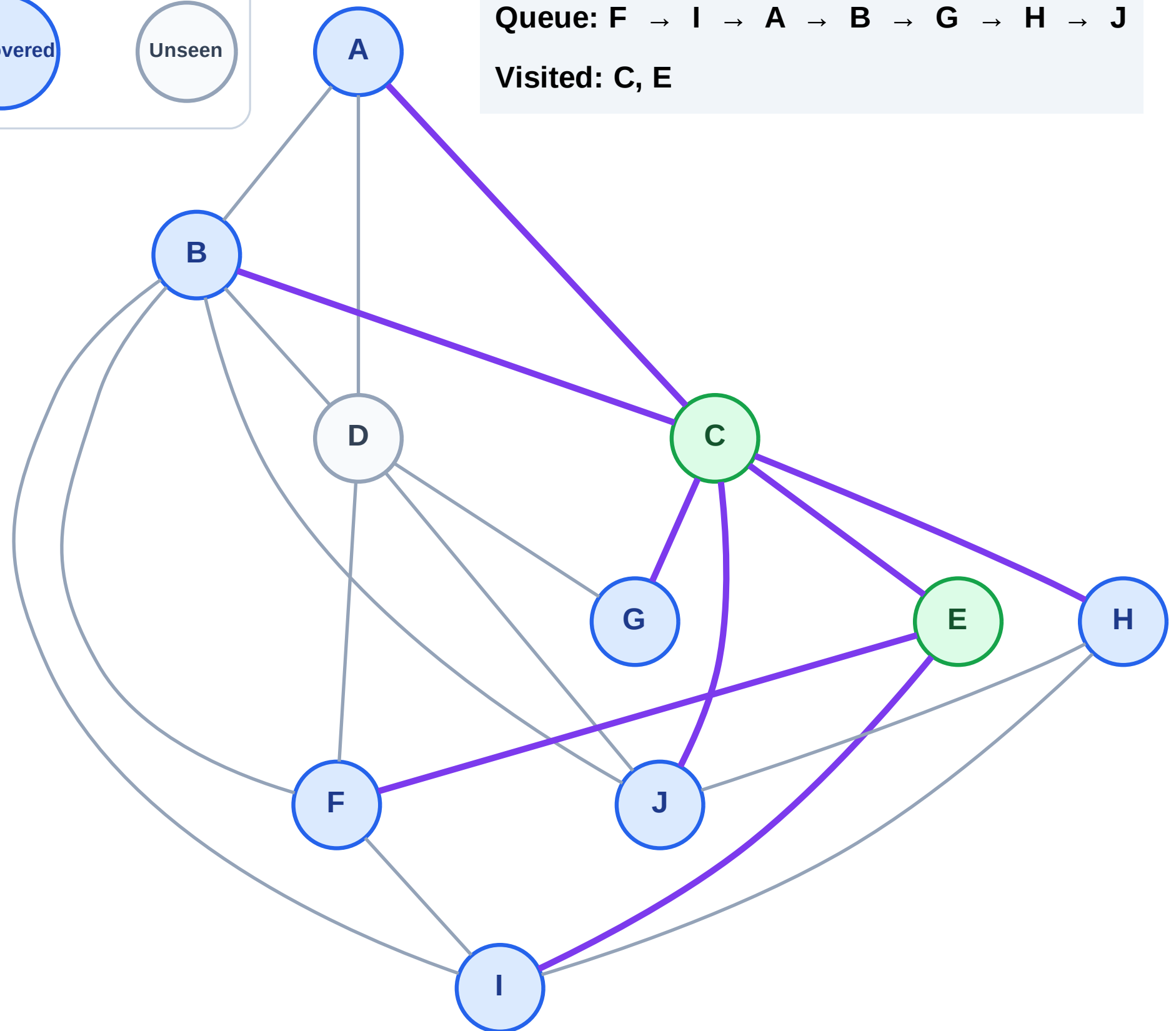
Step 5: Discover A, B, G, H, J from C

Legend



Queue: F → I → A → B → G → H → J

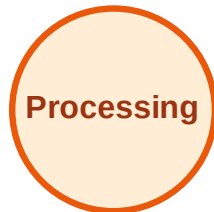
Visited: C, E



BFS Traversal

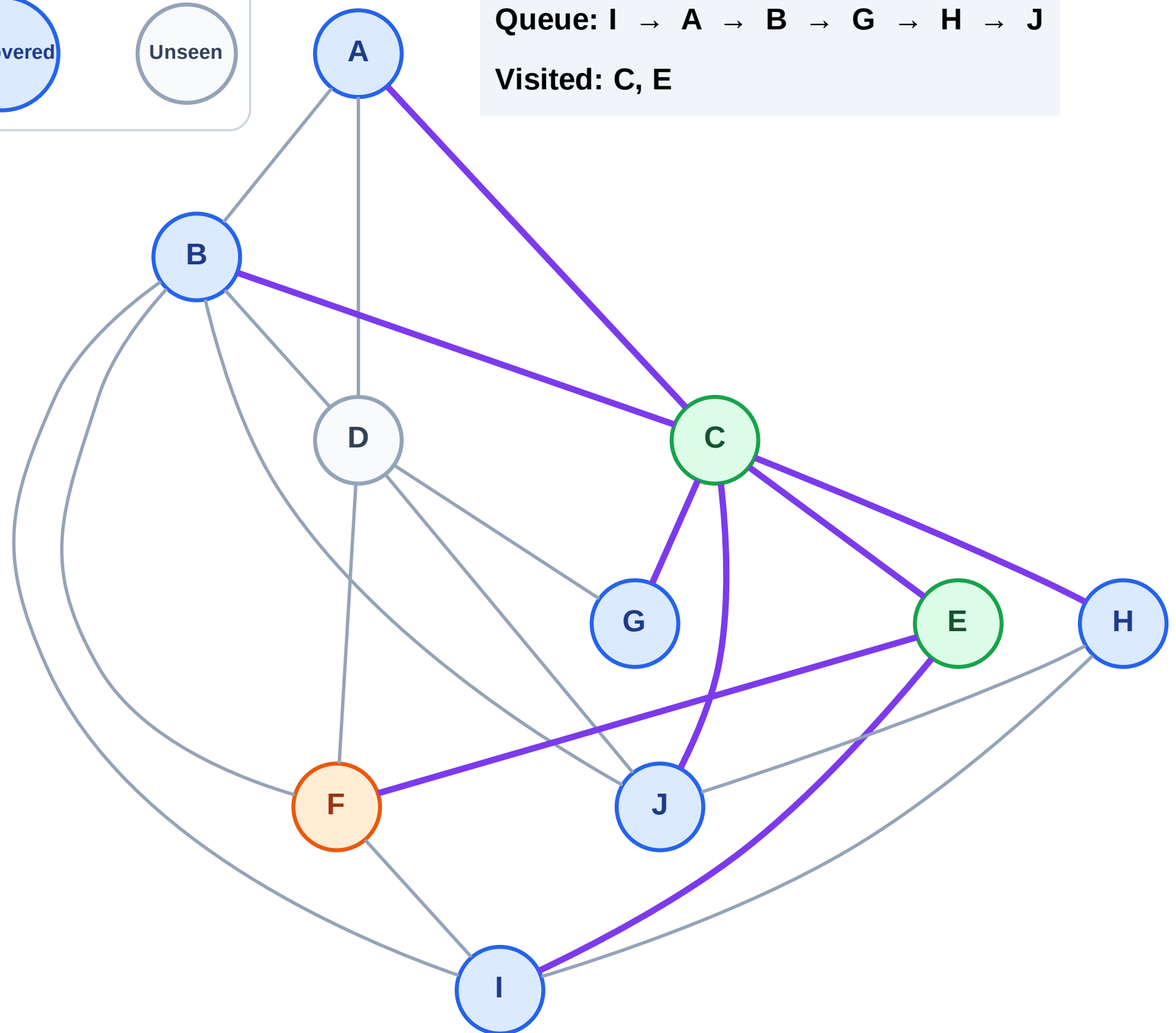
Step 6: Process node F

Legend



Queue: I → A → B → G → H → J

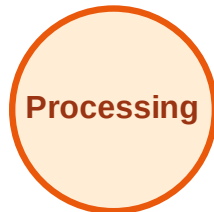
Visited: C, E



BFS Traversal

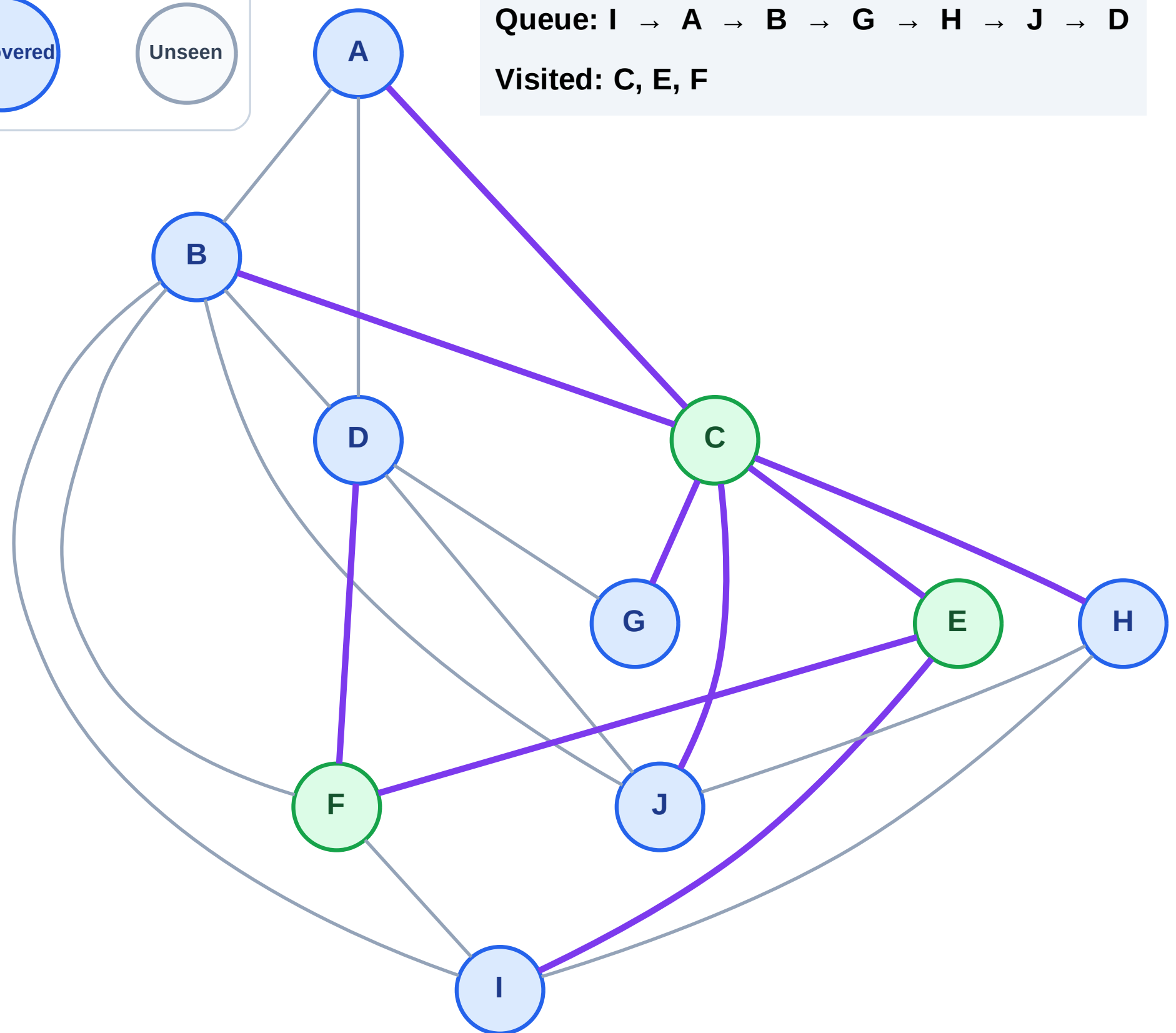
Step 7: Discover D from F

Legend



Queue: I → A → B → G → H → J → D

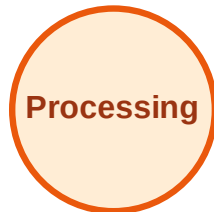
Visited: C, E, F



BFS Traversal

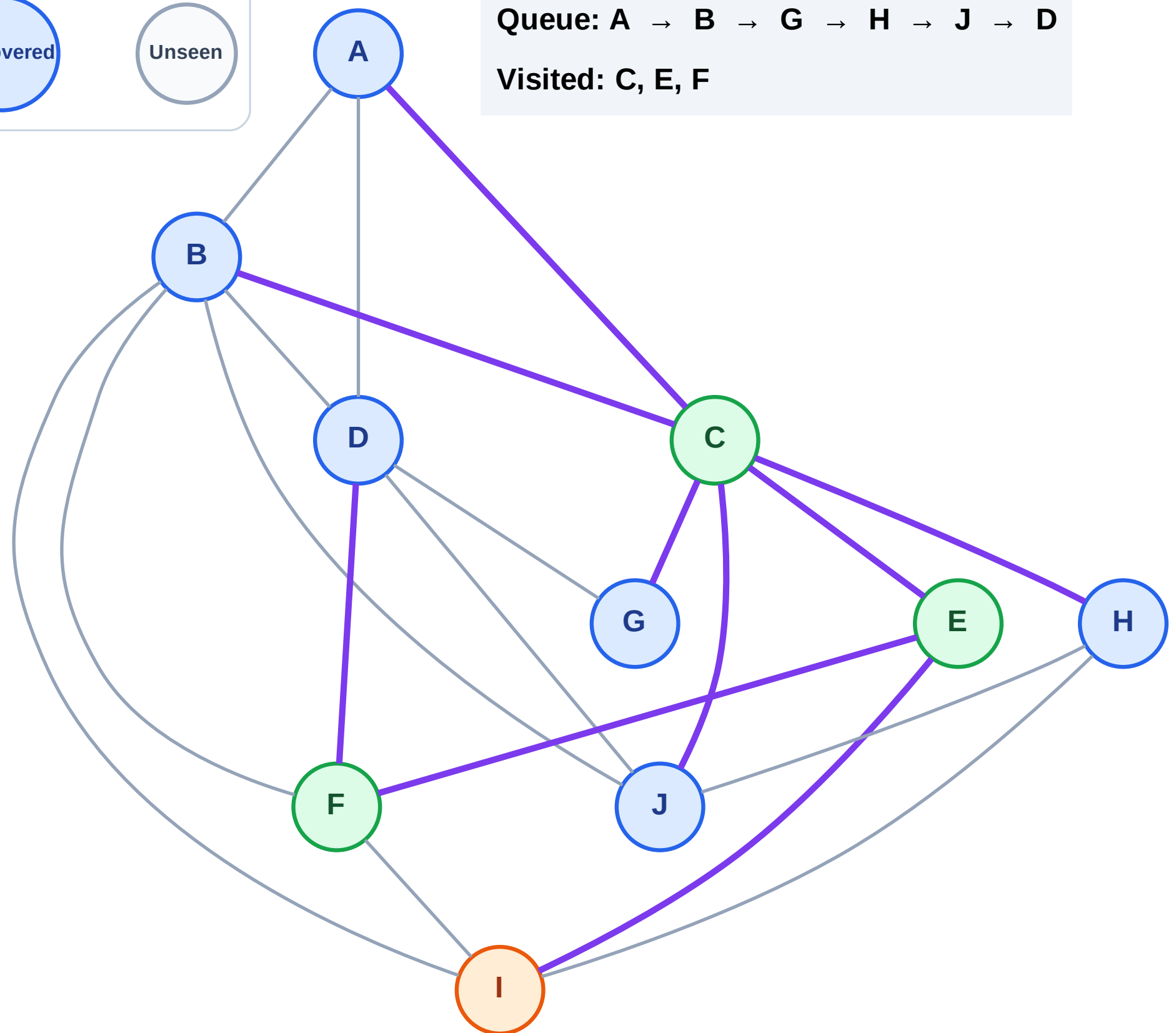
Step 8: Process node I

Legend



Queue: A → B → G → H → J → D

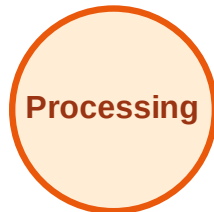
Visited: C, E, F



BFS Traversal

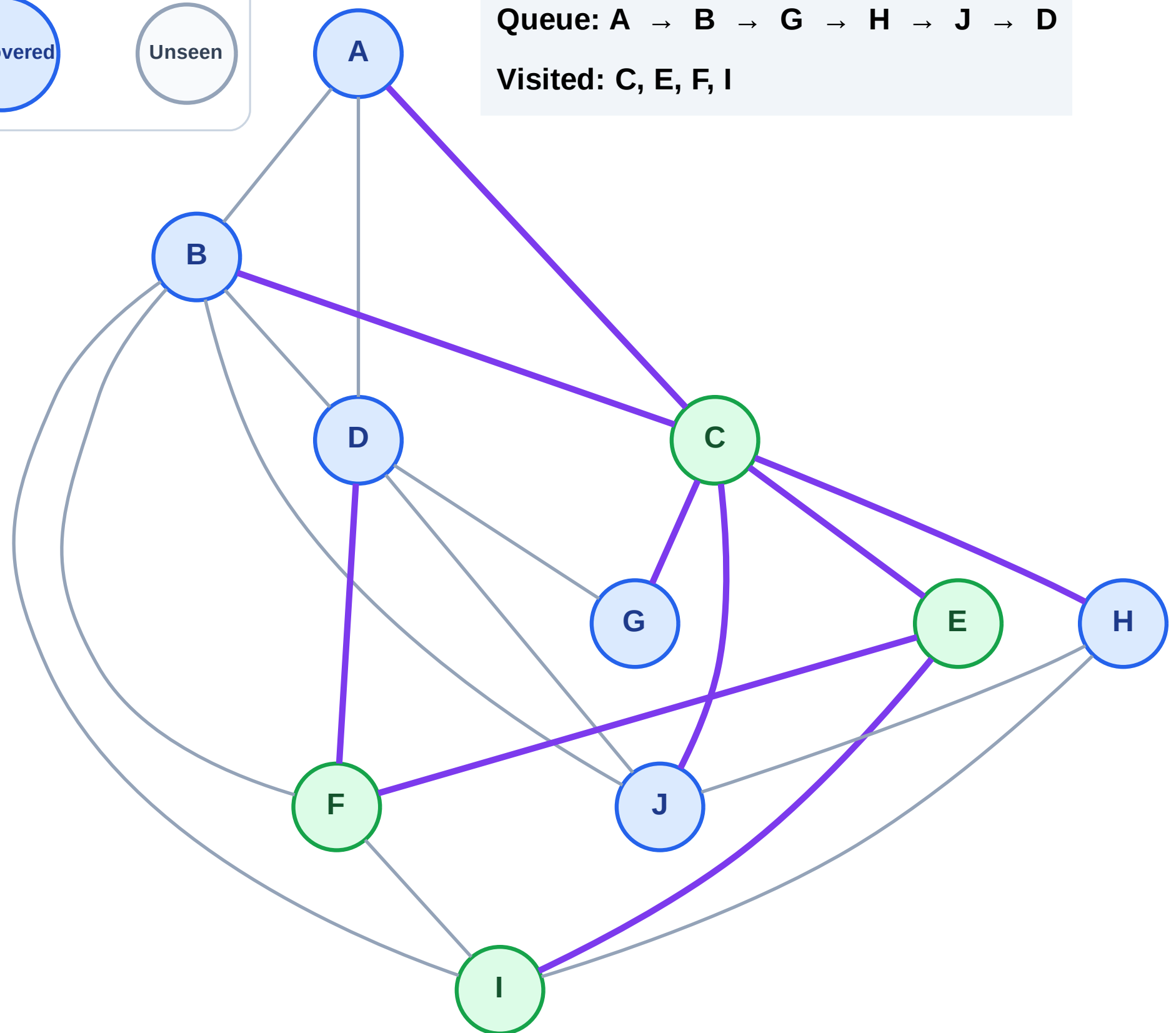
Step 9: No new neighbors from I

Legend



Queue: A → B → G → H → J → D

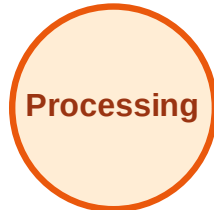
Visited: C, E, F, I



BFS Traversal

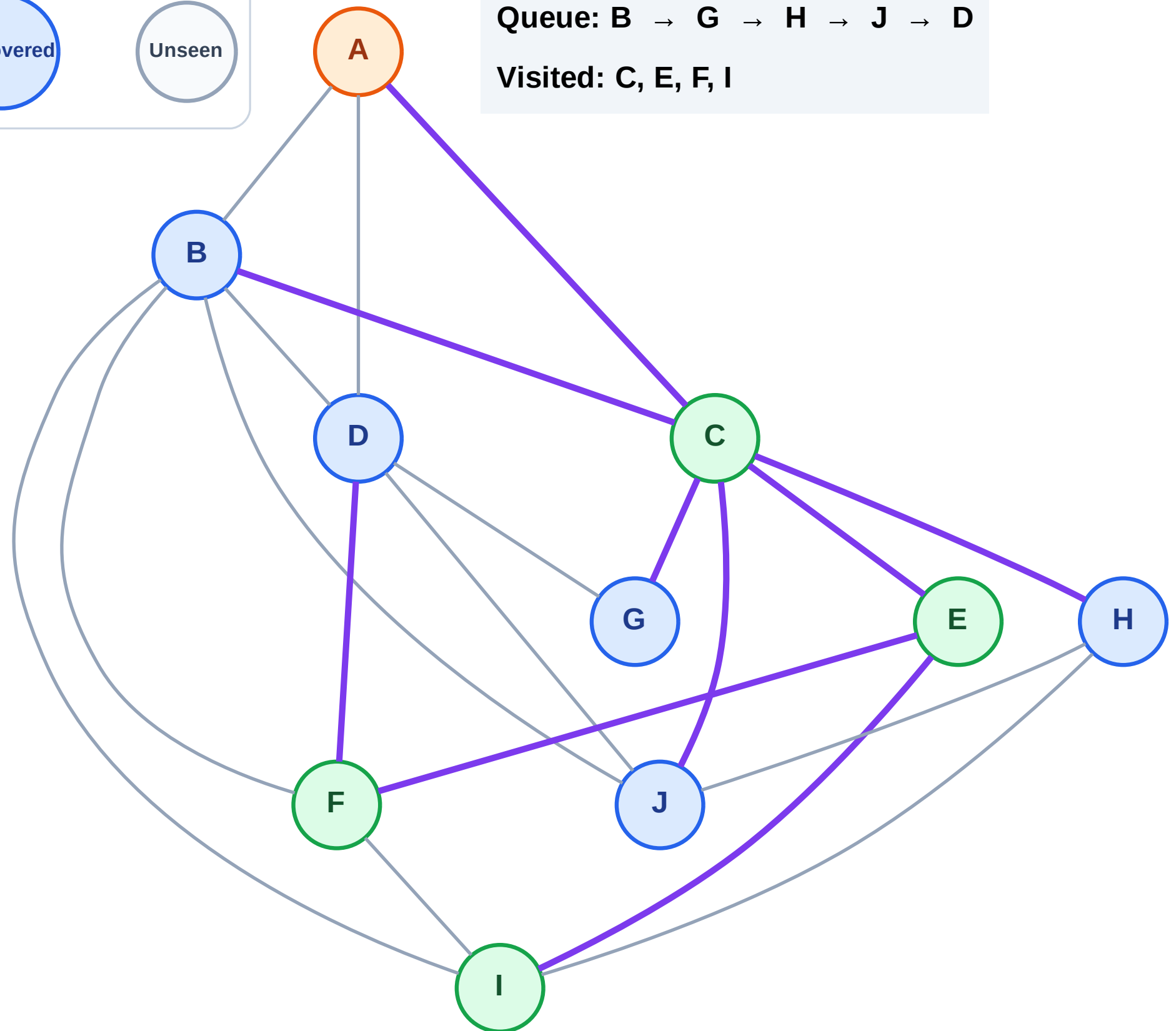
Step 10: Process node A

Legend



Queue: B → G → H → J → D

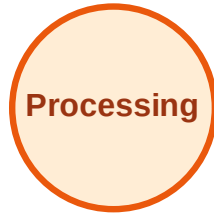
Visited: C, E, F, I



BFS Traversal

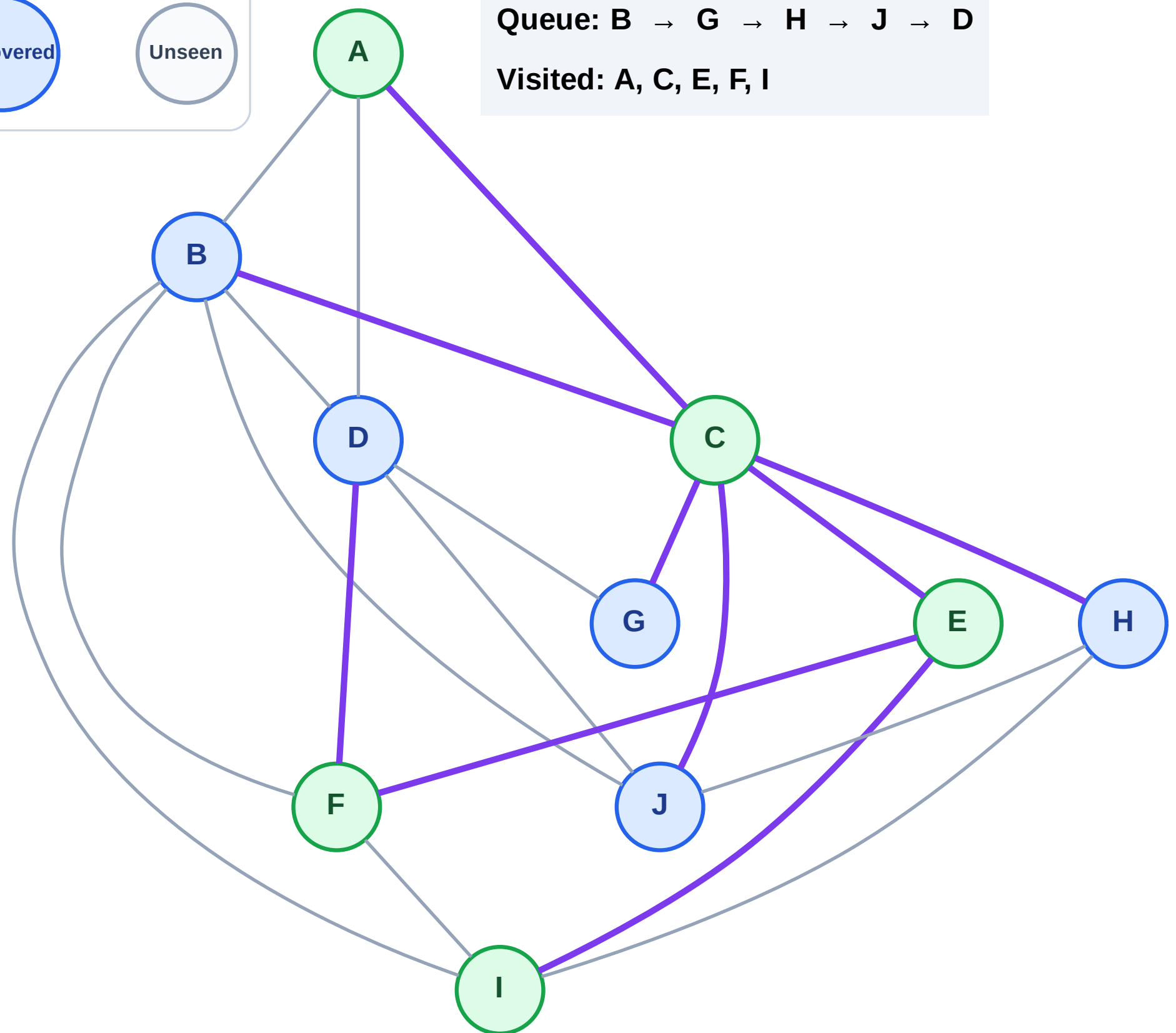
Step 11: No new neighbors from A

Legend



Queue: B → G → H → J → D

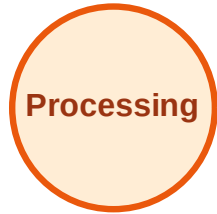
Visited: A, C, E, F, I



BFS Traversal

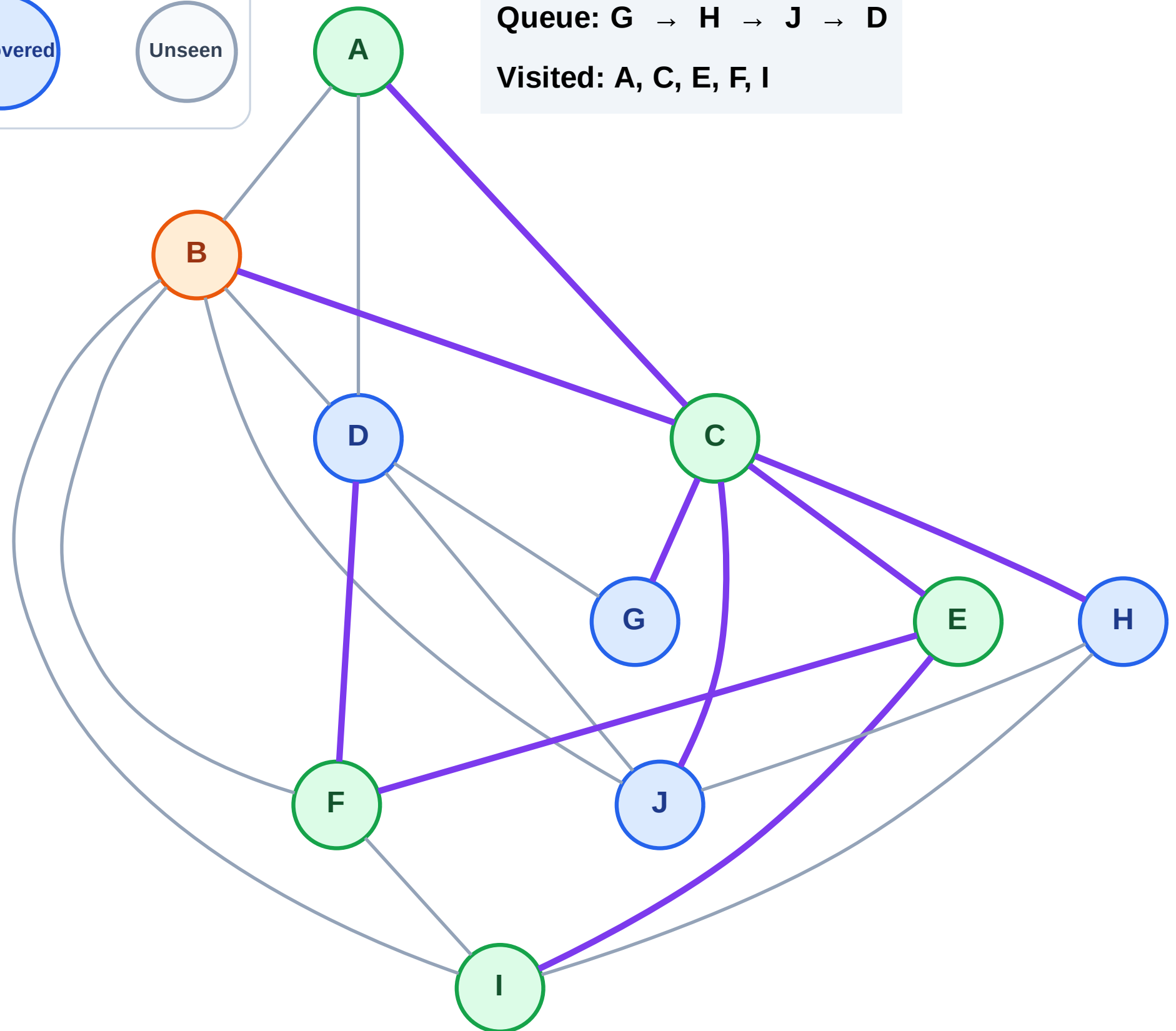
Step 12: Process node B

Legend



Queue: G → H → J → D

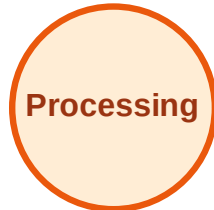
Visited: A, C, E, F, I



BFS Traversal

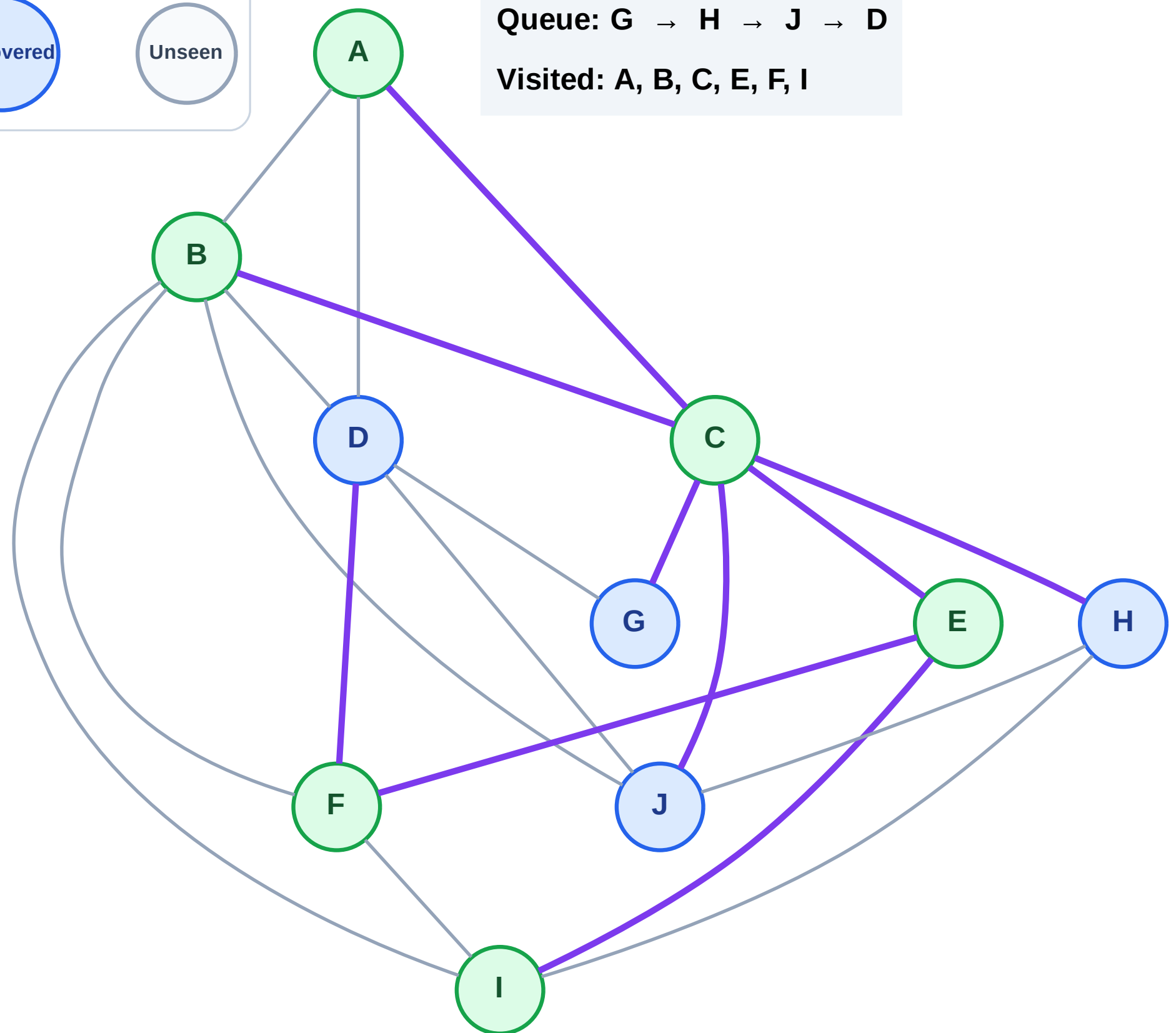
Step 13: No new neighbors from B

Legend



Queue: G → H → J → D

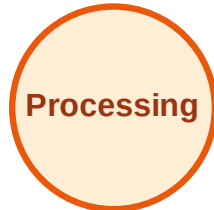
Visited: A, B, C, E, F, I



BFS Traversal

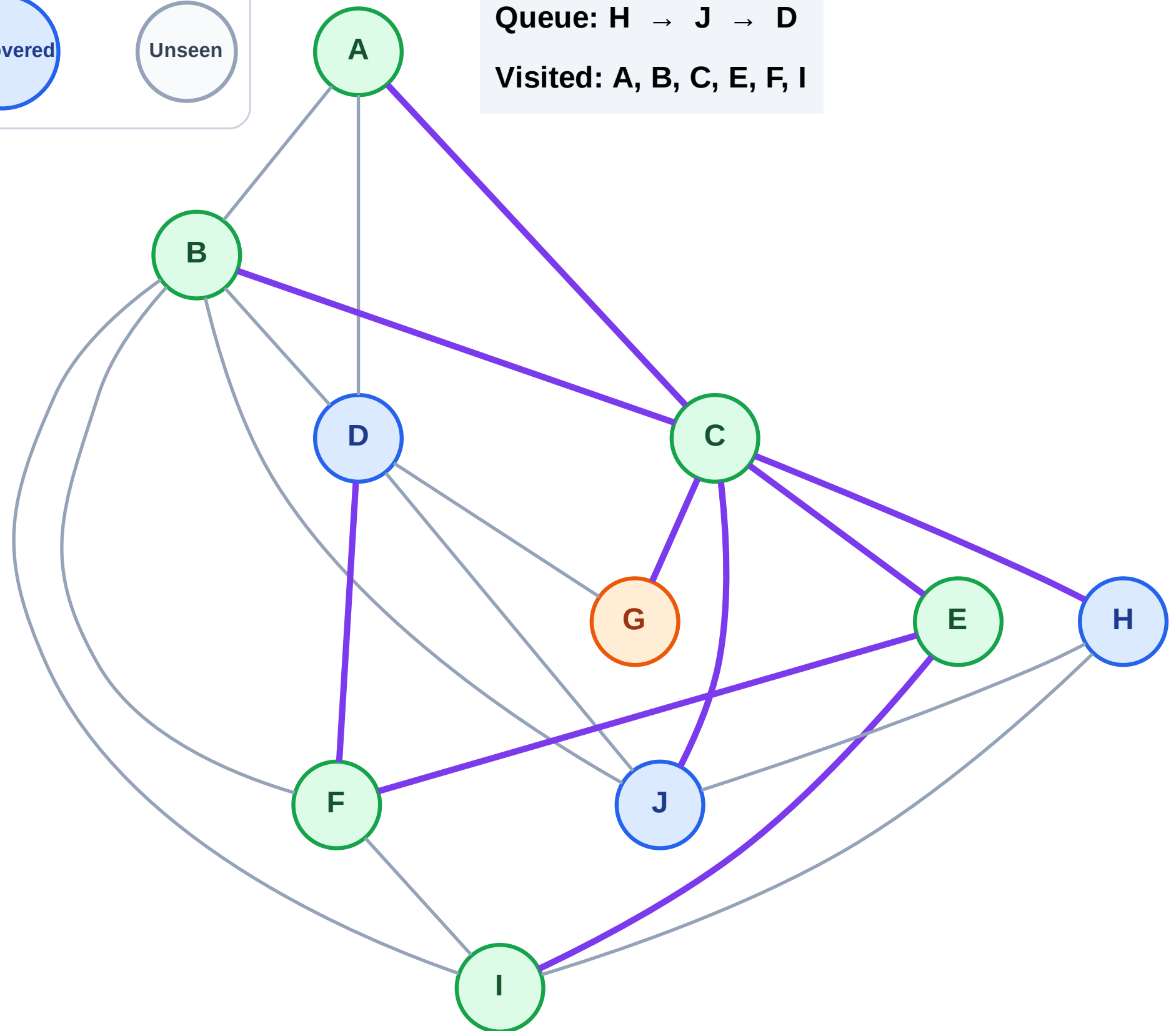
Step 14: Process node G

Legend



Queue: H → J → D

Visited: A, B, C, E, F, I



BFS Traversal

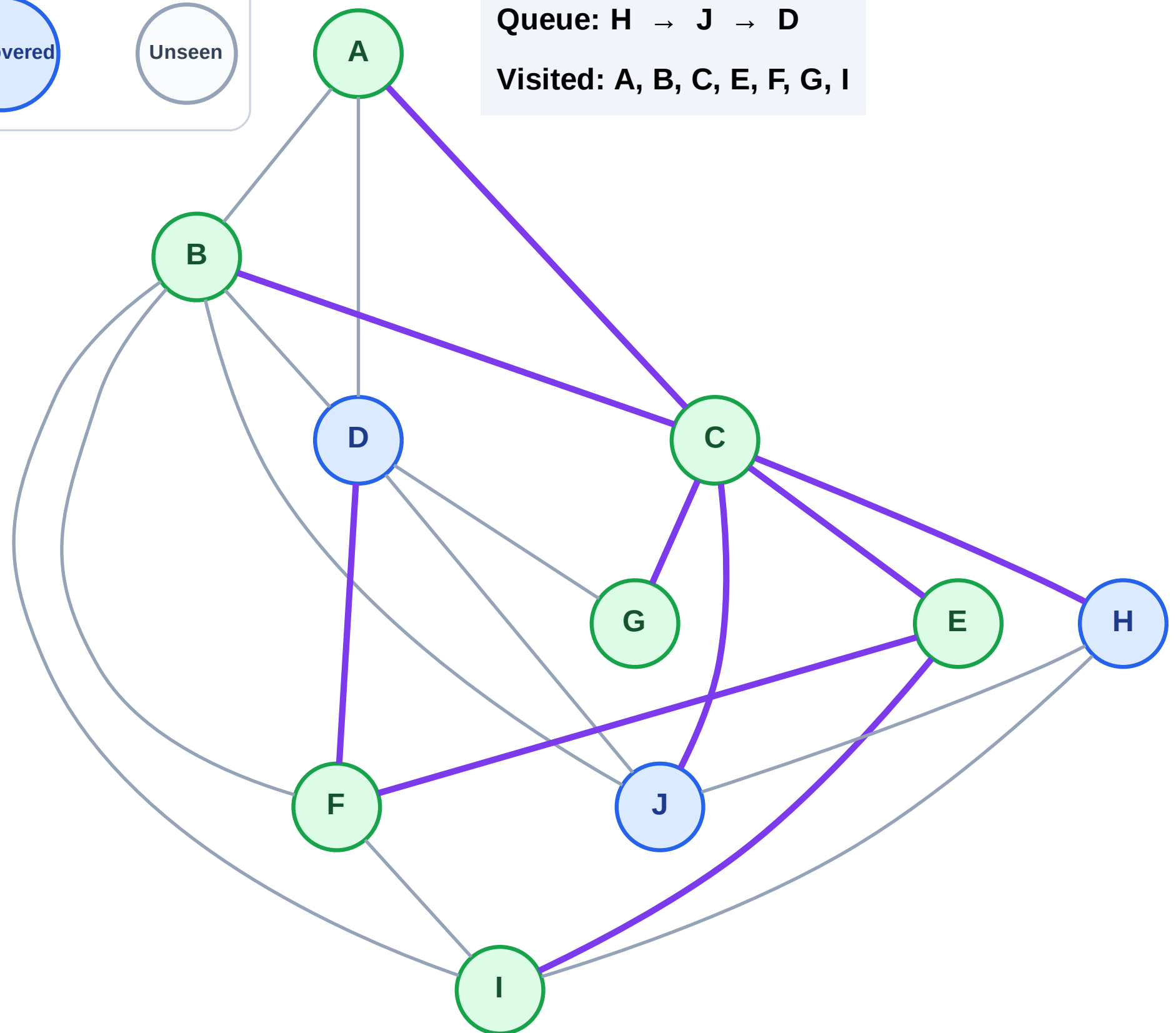
Step 15: No new neighbors from G

Legend



Queue: H → J → D

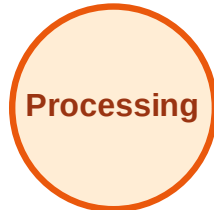
Visited: A, B, C, E, F, G, I



BFS Traversal

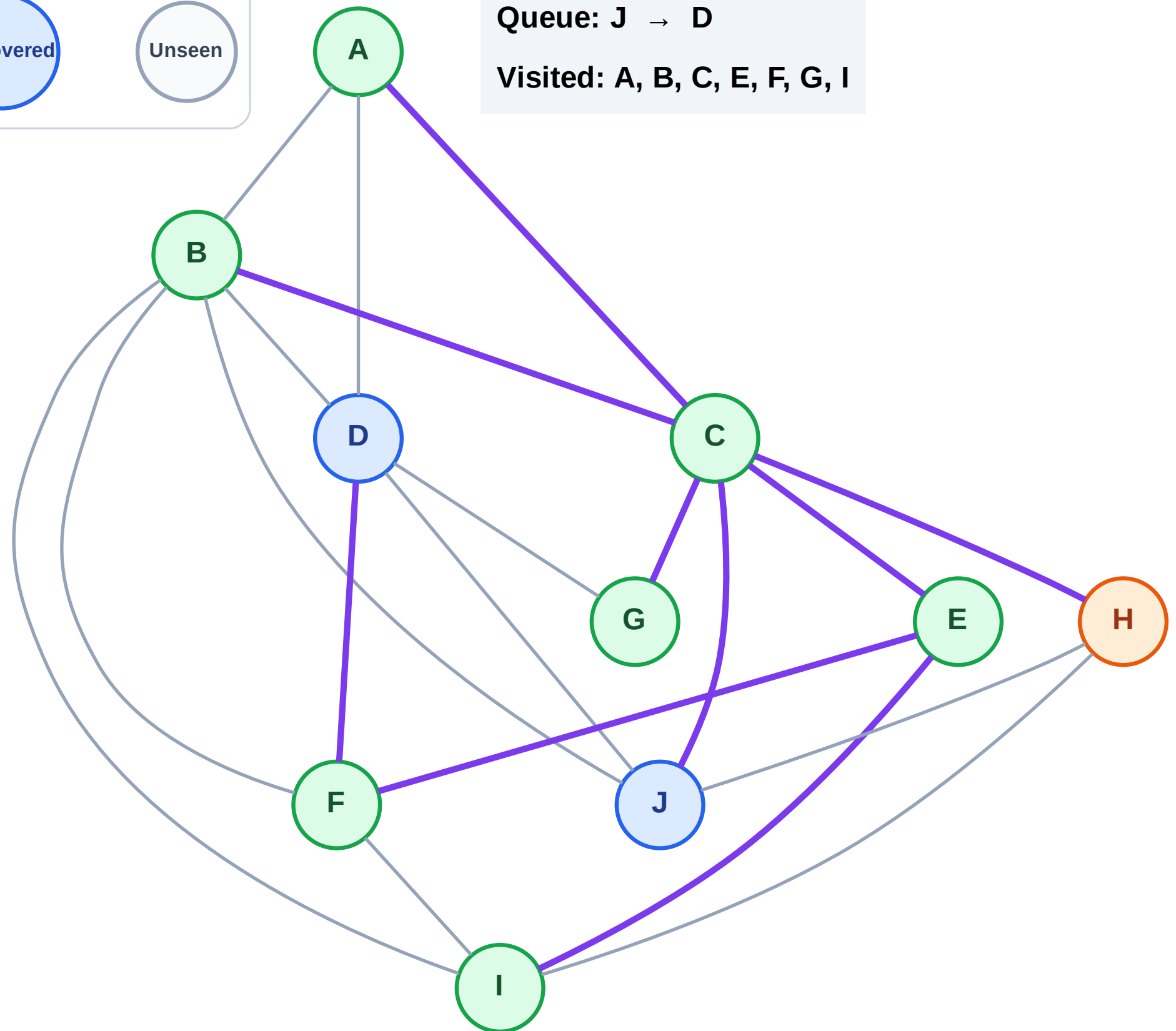
Step 16: Process node H

Legend



Queue: J → D

Visited: A, B, C, E, F, G, I



BFS Traversal

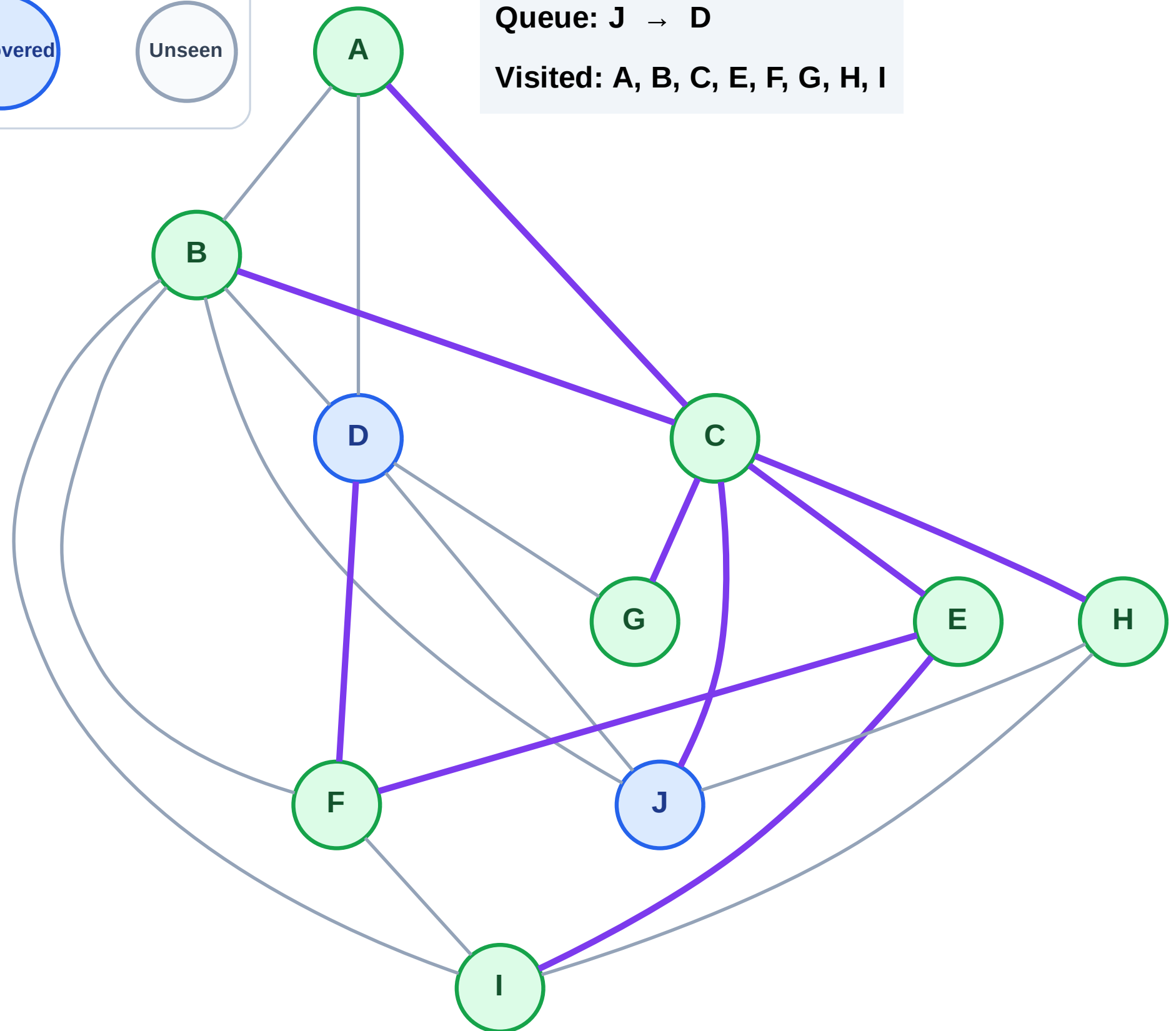
Step 17: No new neighbors from H

Legend



Queue: J → D

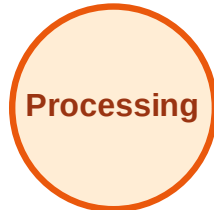
Visited: A, B, C, E, F, G, H, I



BFS Traversal

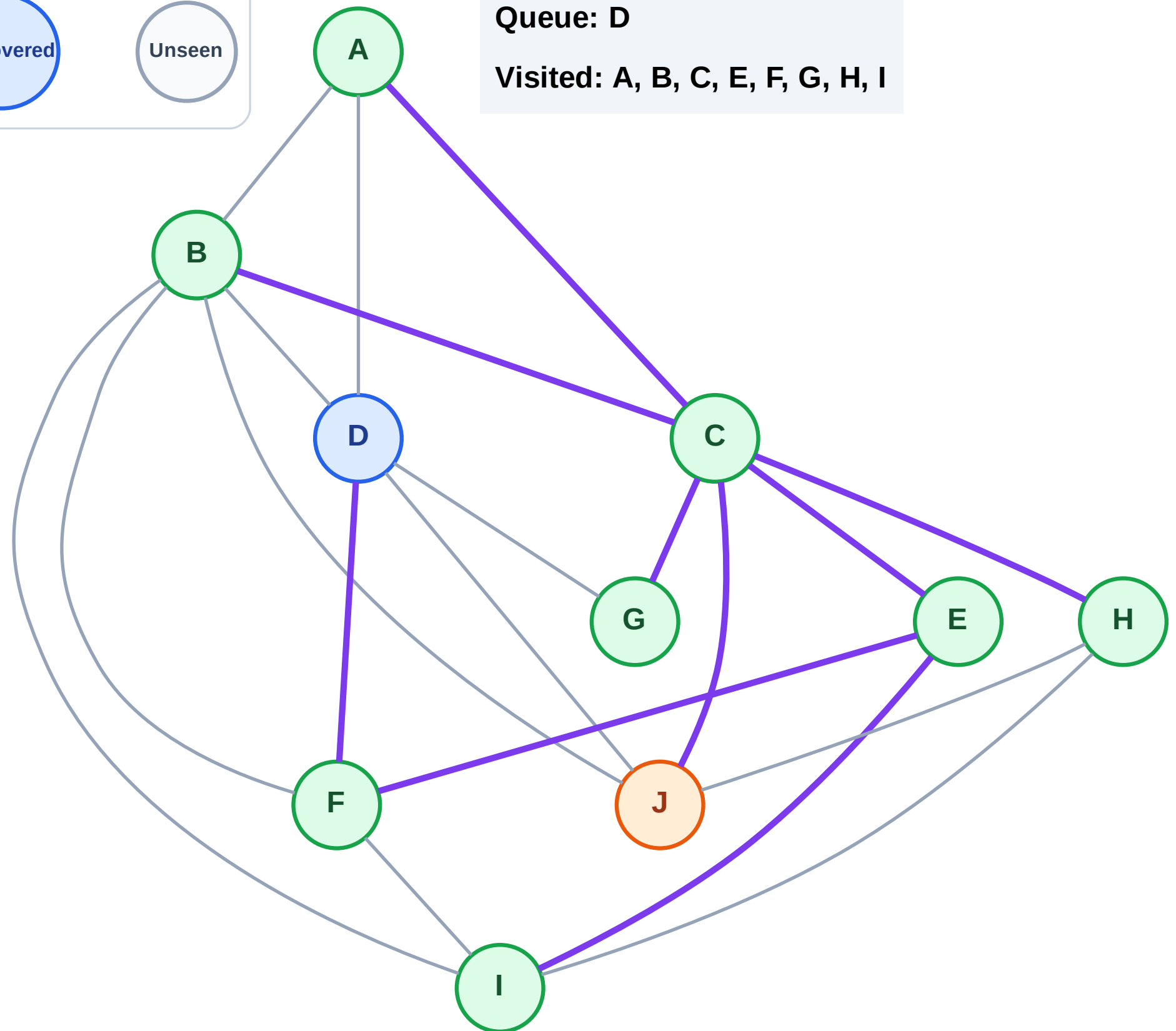
Step 18: Process node J

Legend



Queue: D

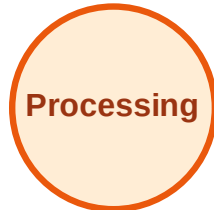
Visited: A, B, C, E, F, G, H, I



BFS Traversal

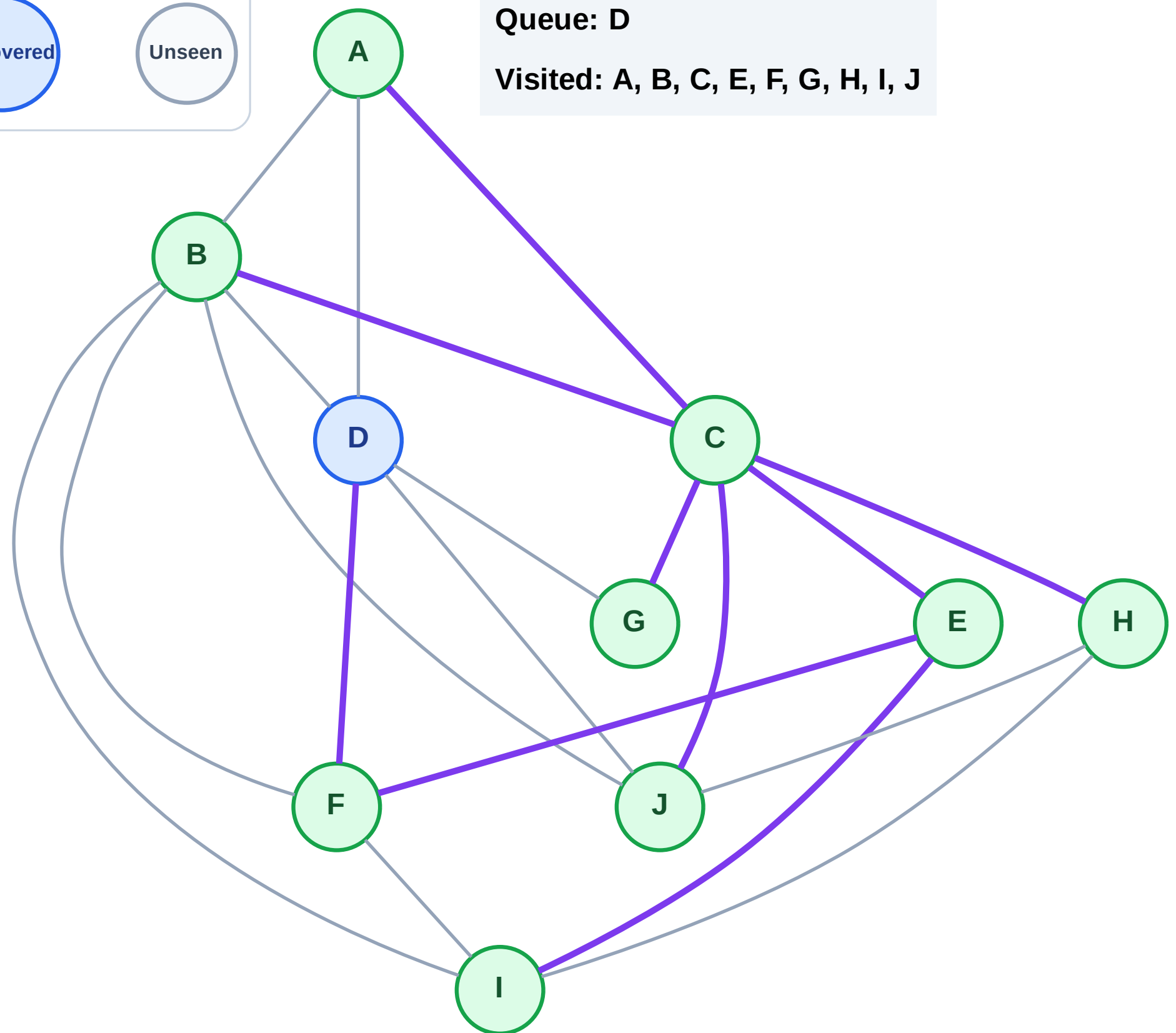
Step 19: No new neighbors from J

Legend



Queue: D

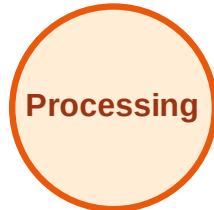
Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

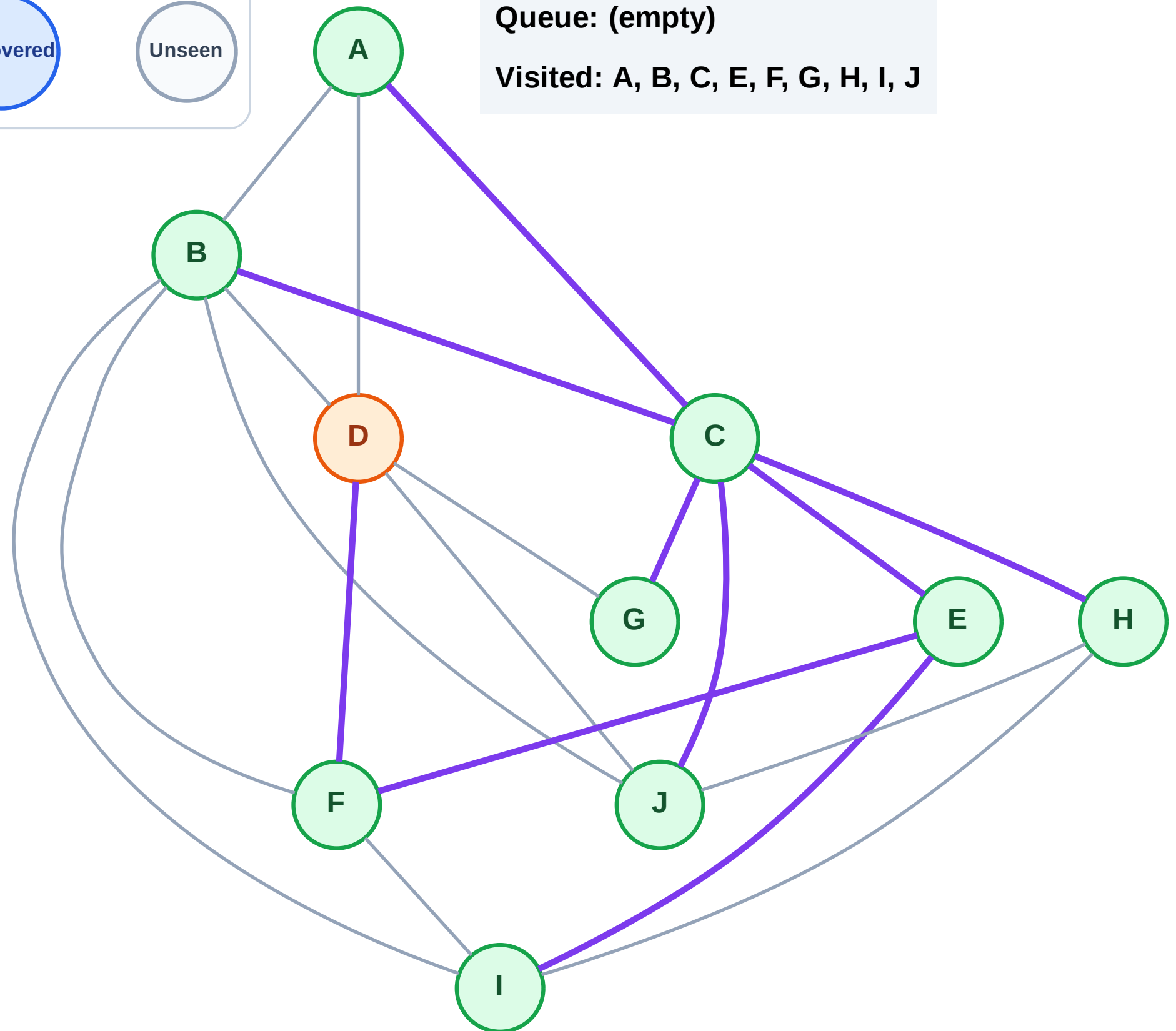
Step 20: Process node D

Legend



Queue: (empty)

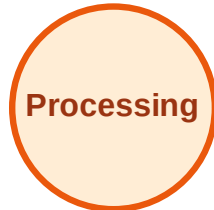
Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

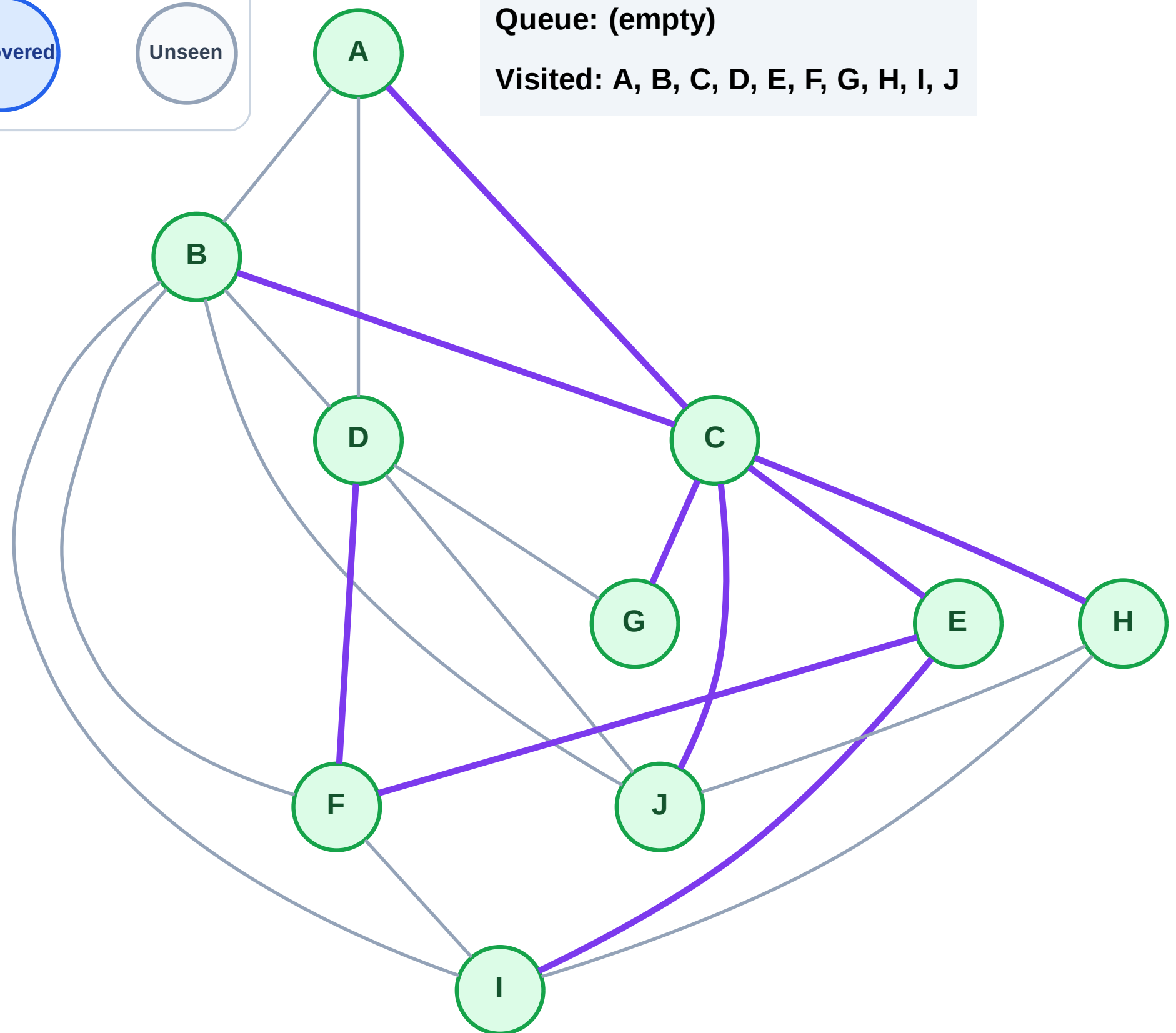
Step 21: No new neighbors from D

Legend



Queue: (empty)

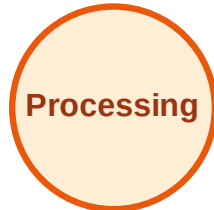
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

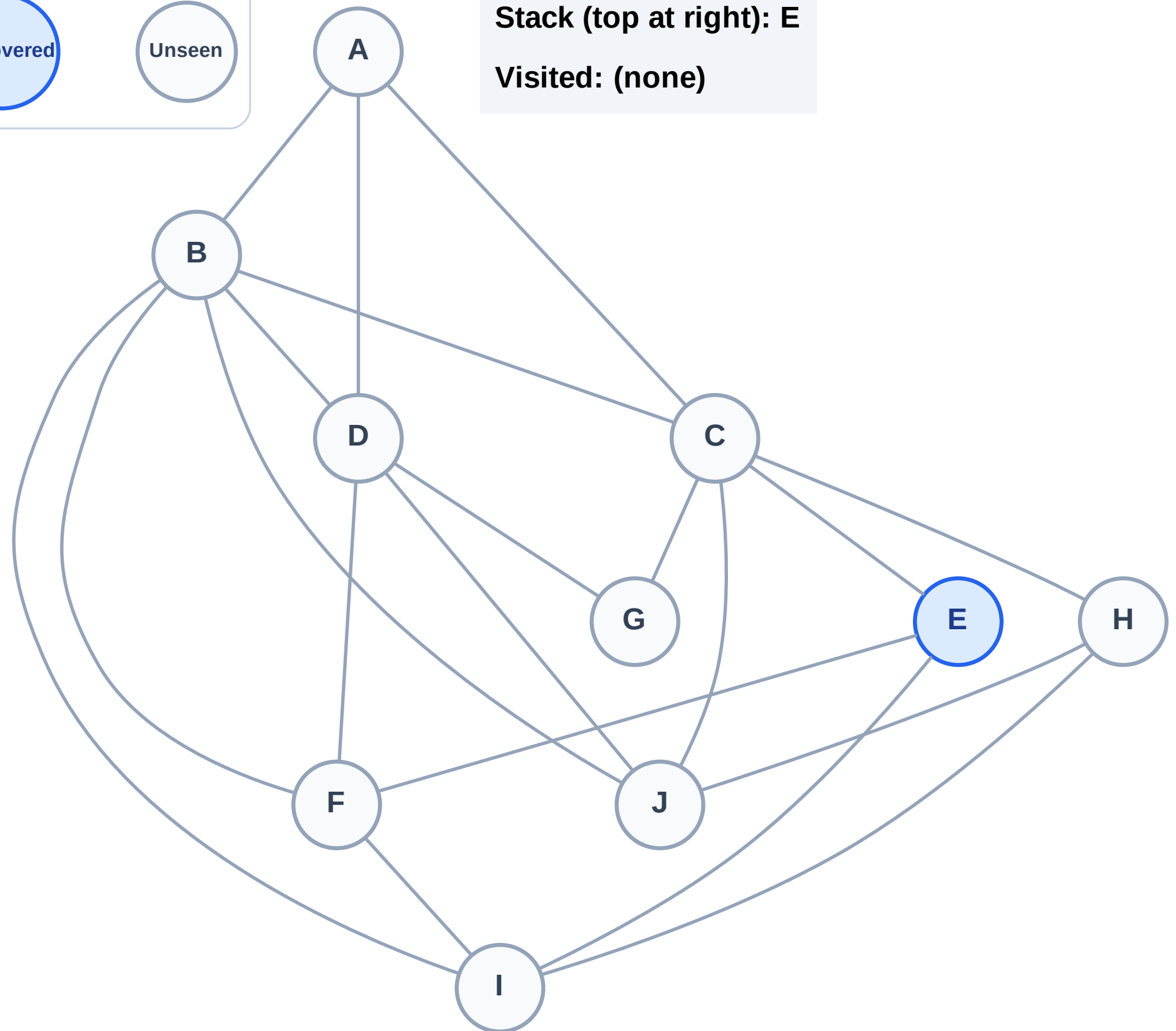
Step 1: Start at E

Legend



Stack (top at right): E

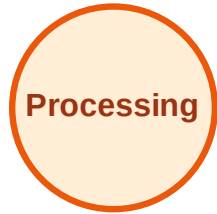
Visited: (none)



DFS Traversal

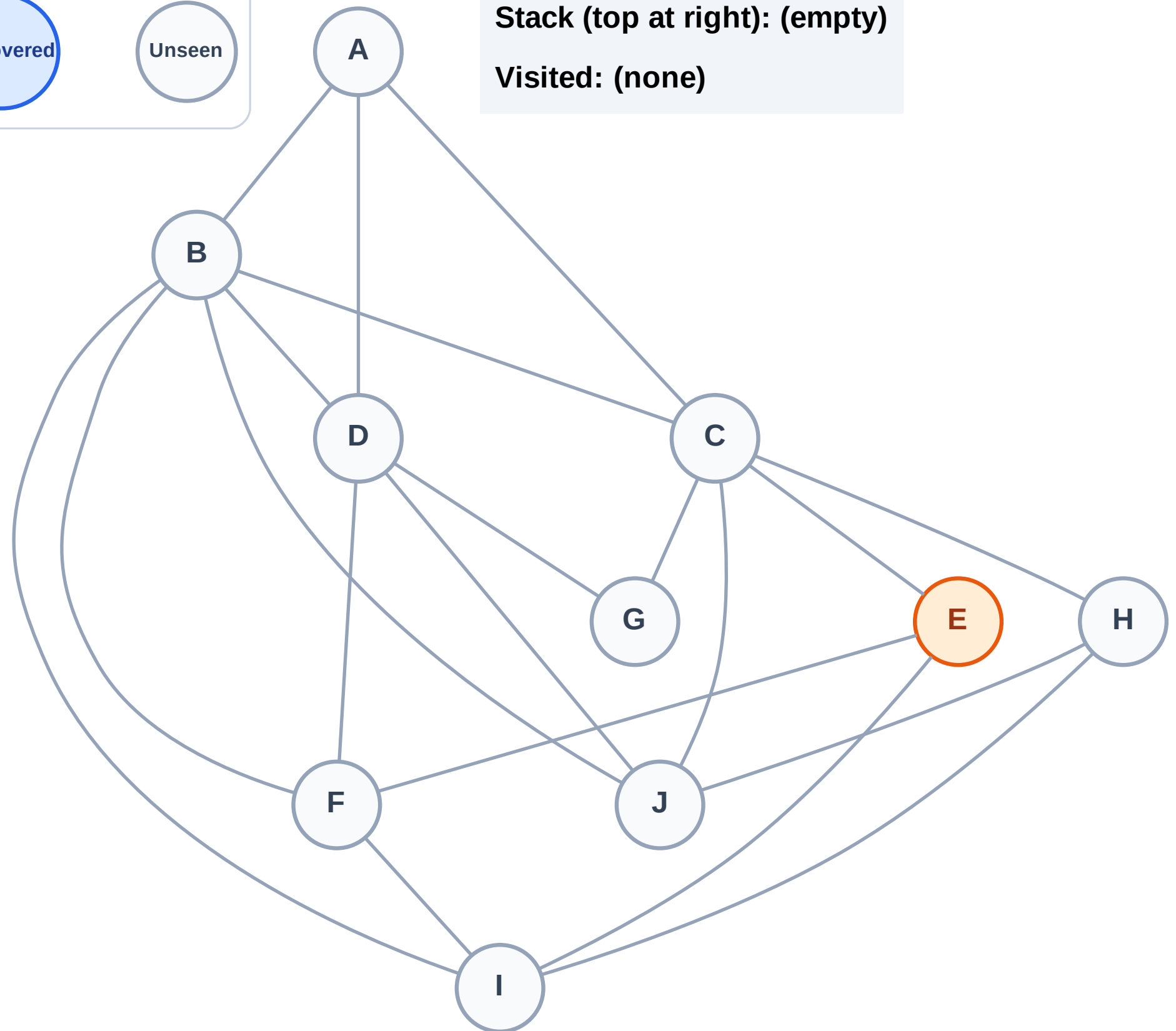
Step 2: Process node E

Legend



Stack (top at right): (empty)

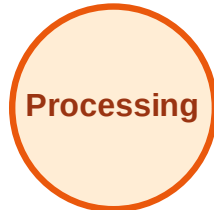
Visited: (none)



DFS Traversal

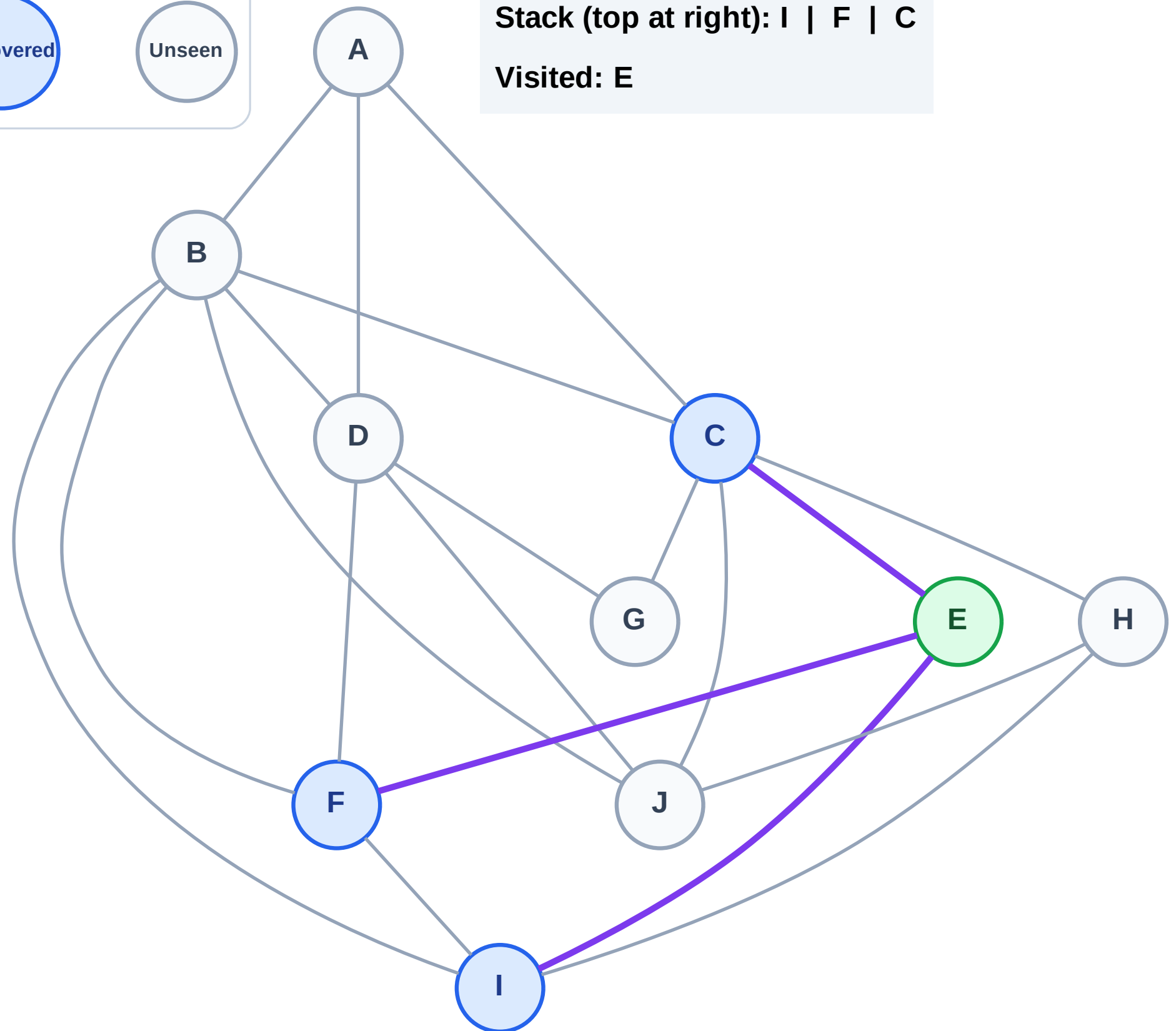
Step 3: Discover I, F, C from E

Legend



Stack (top at right): I | F | C

Visited: E



DFS Traversal

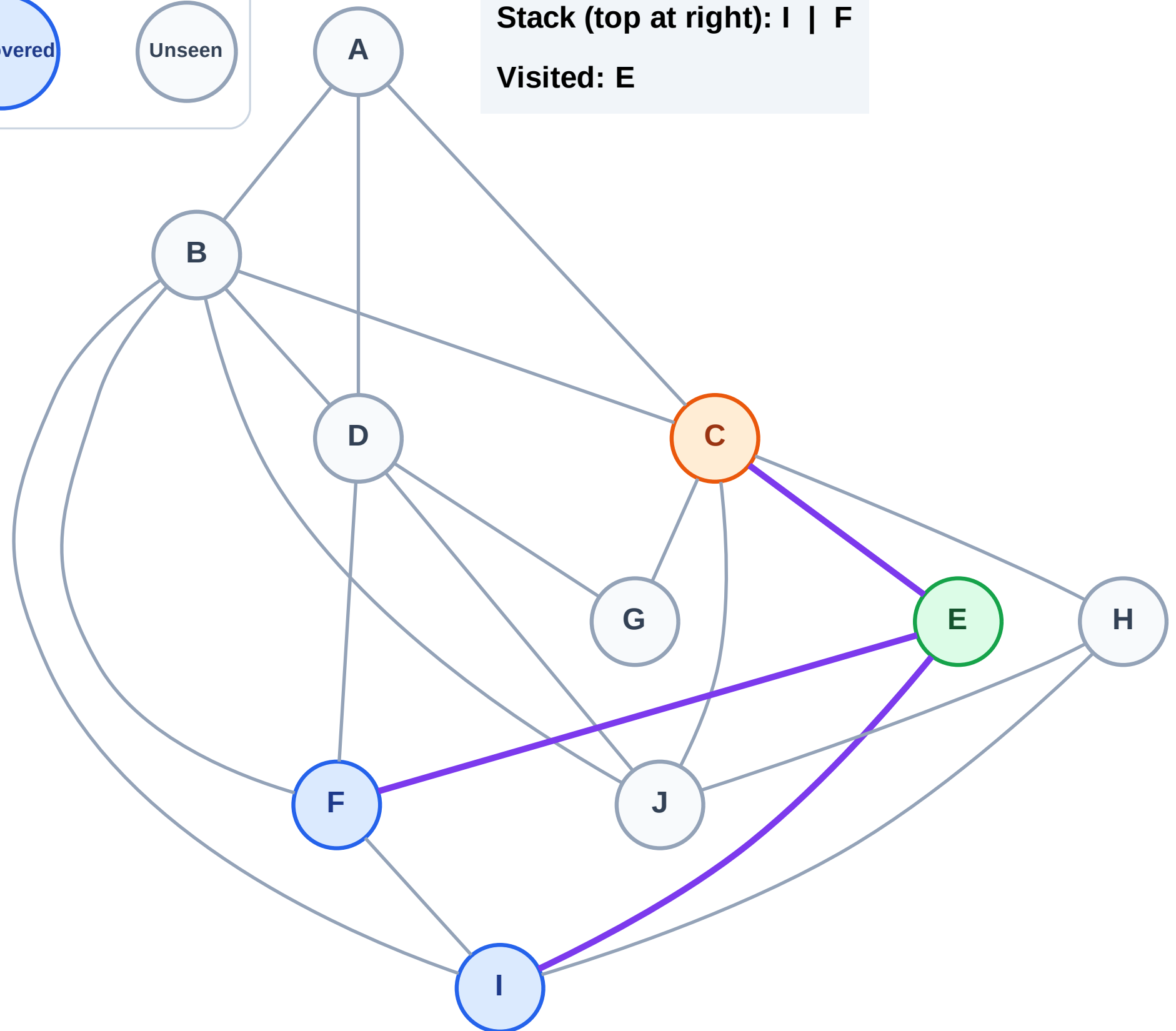
Step 4: Process node C

Legend



Stack (top at right): I | F

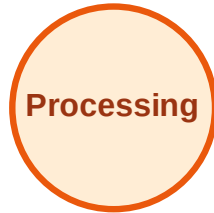
Visited: E



DFS Traversal

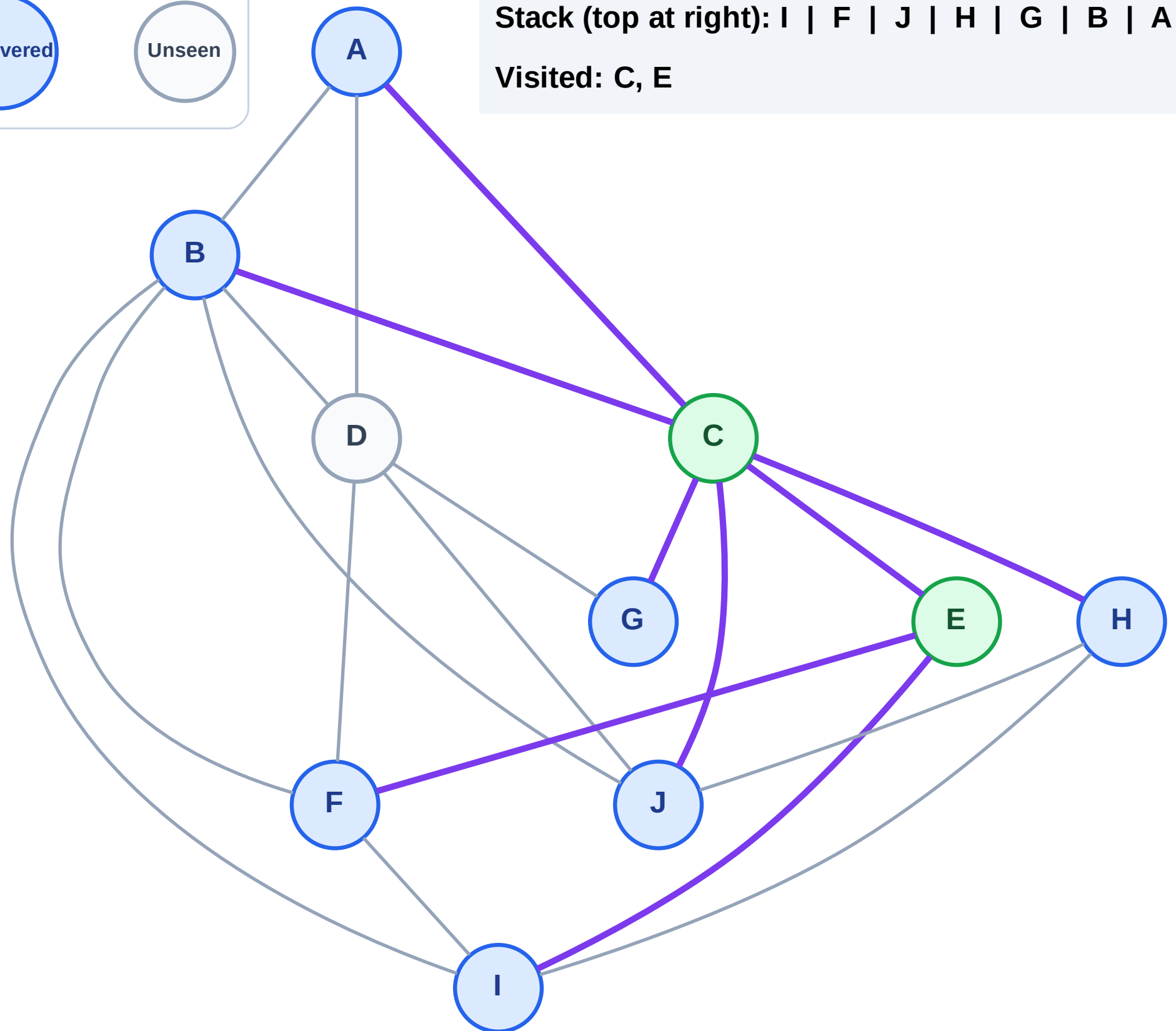
Step 5: Discover J, H, G, B, A from C

Legend



Stack (top at right): I | F | J | H | G | B | A

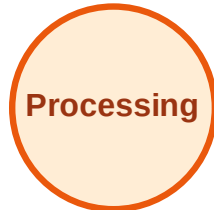
Visited: C, E



DFS Traversal

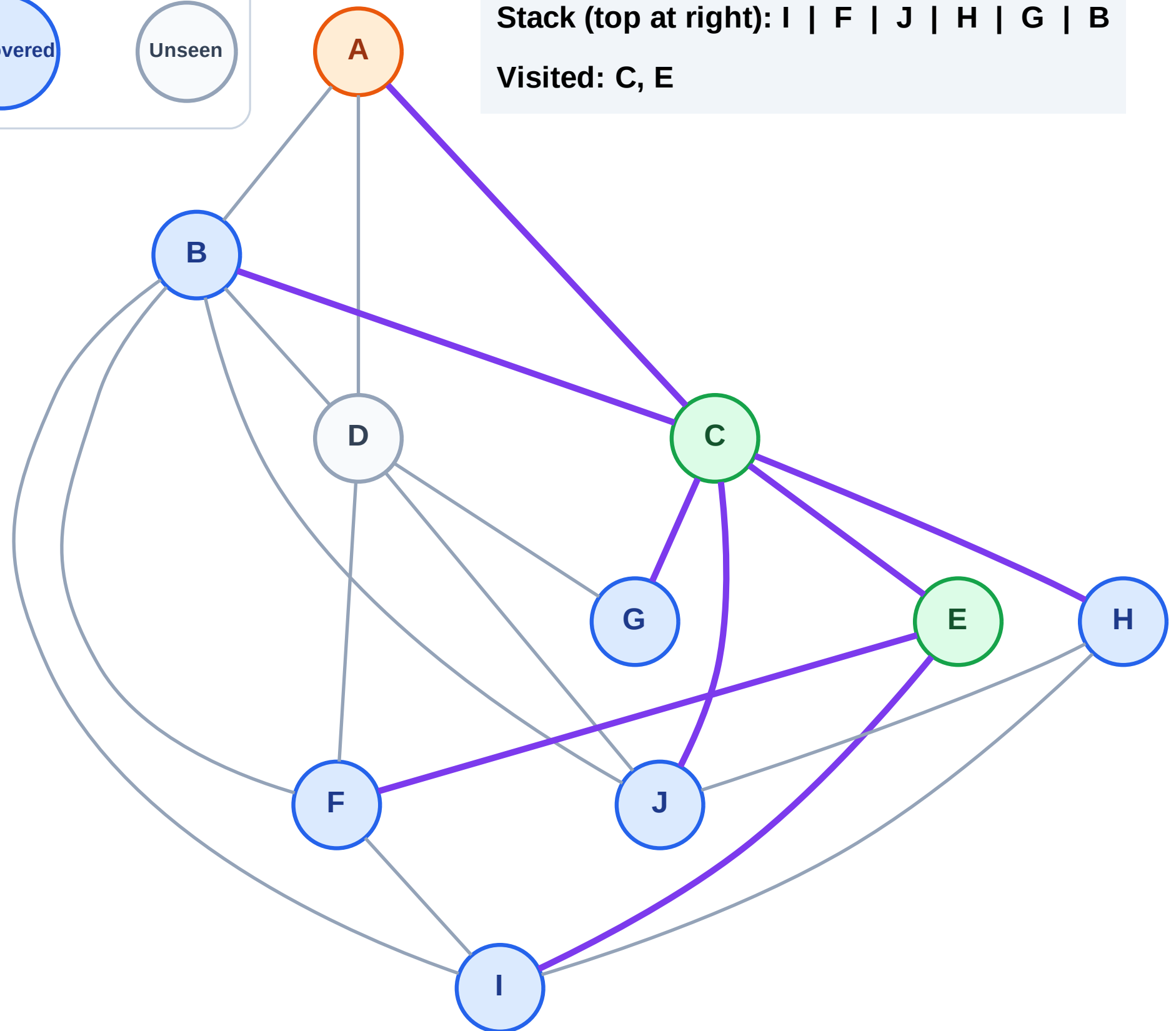
Step 6: Process node A

Legend



Stack (top at right): I | F | J | H | G | B

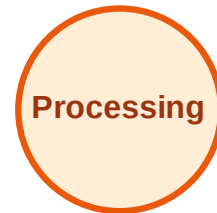
Visited: C, E



DFS Traversal

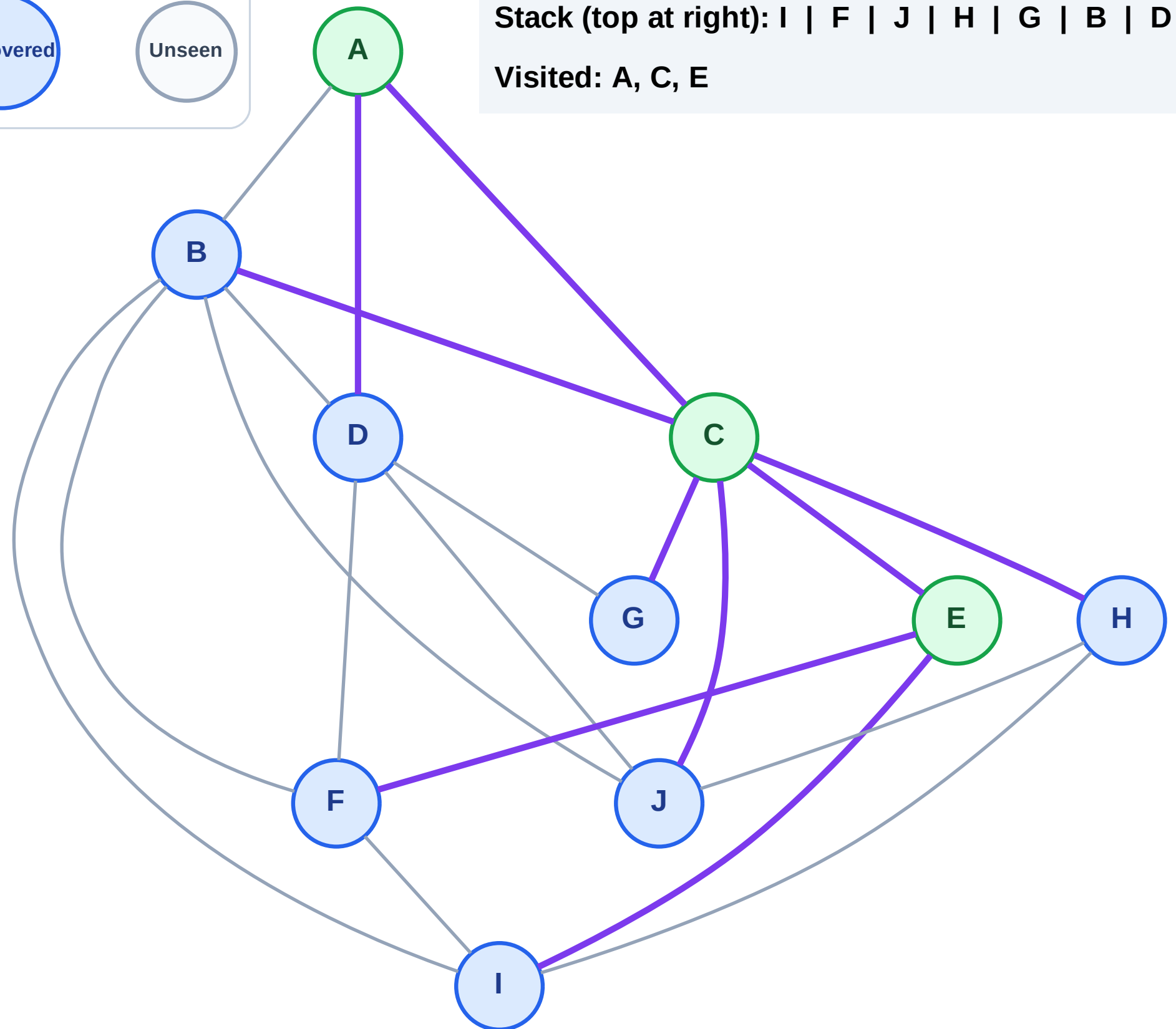
Step 7: Discover D from A

Legend



Stack (top at right): I | F | J | H | G | B | D

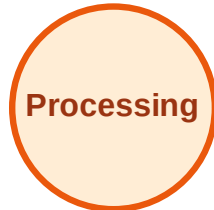
Visited: A, C, E



DFS Traversal

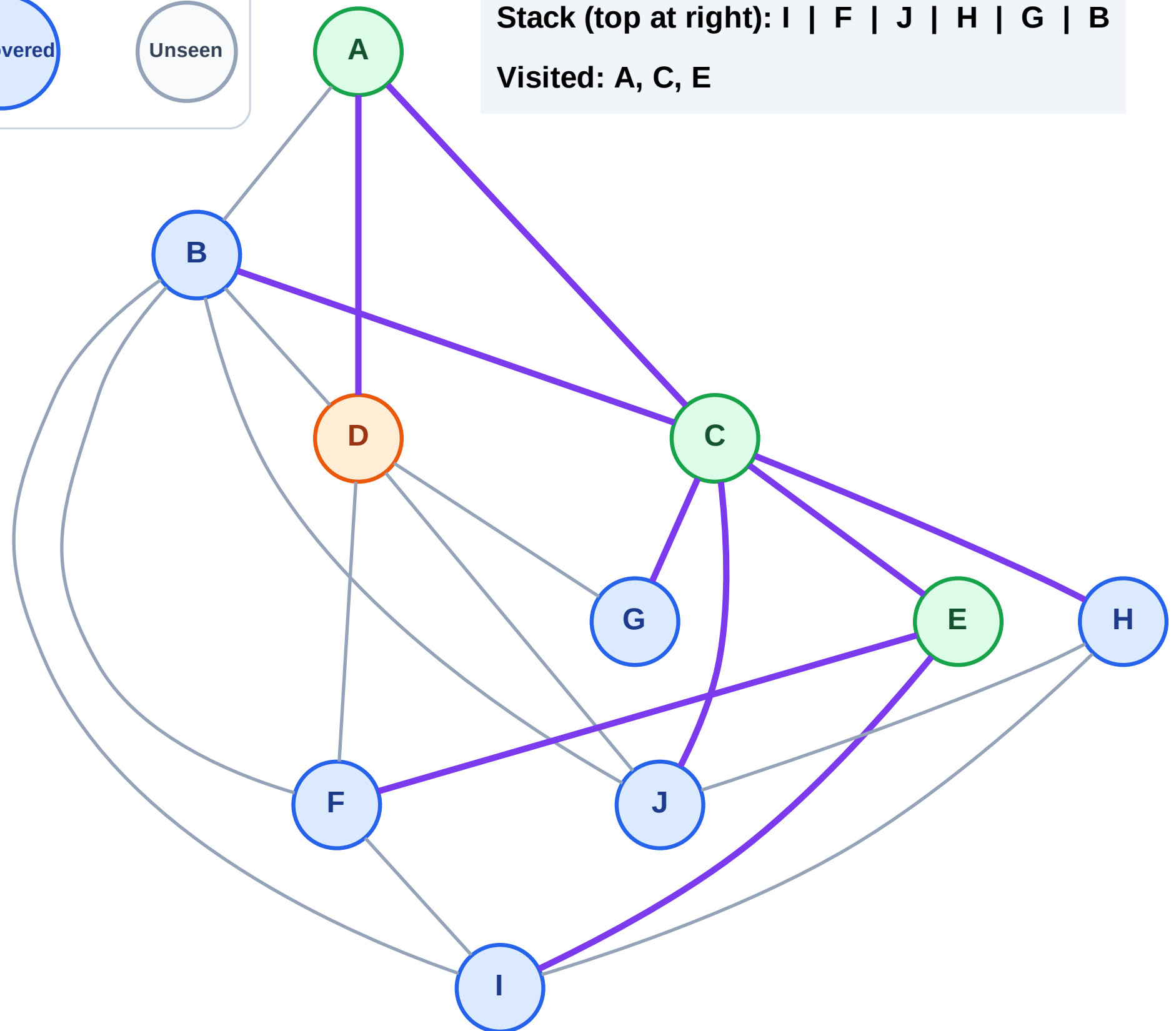
Step 8: Process node D

Legend



Stack (top at right): I | F | J | H | G | B

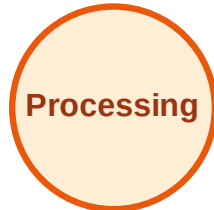
Visited: A, C, E



DFS Traversal

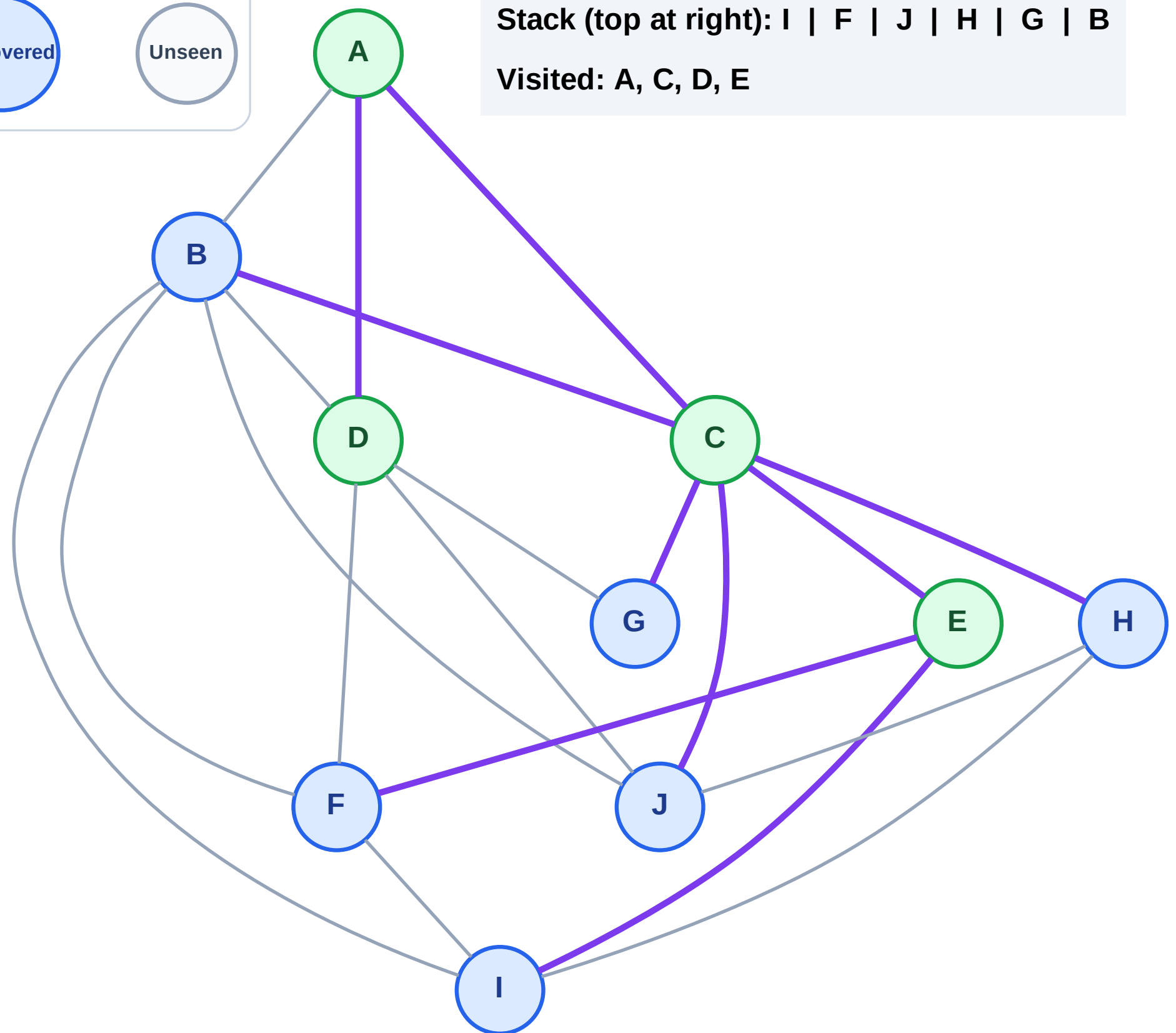
Step 9: No new neighbors from D

Legend



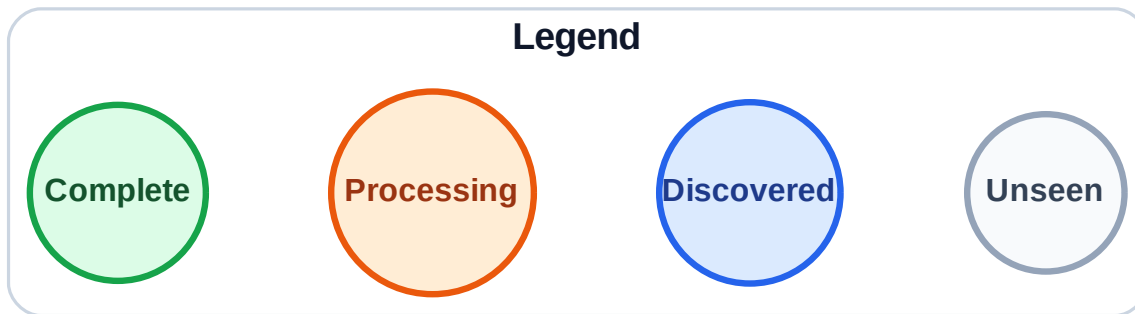
Stack (top at right): I | F | J | H | G | B

Visited: A, C, D, E



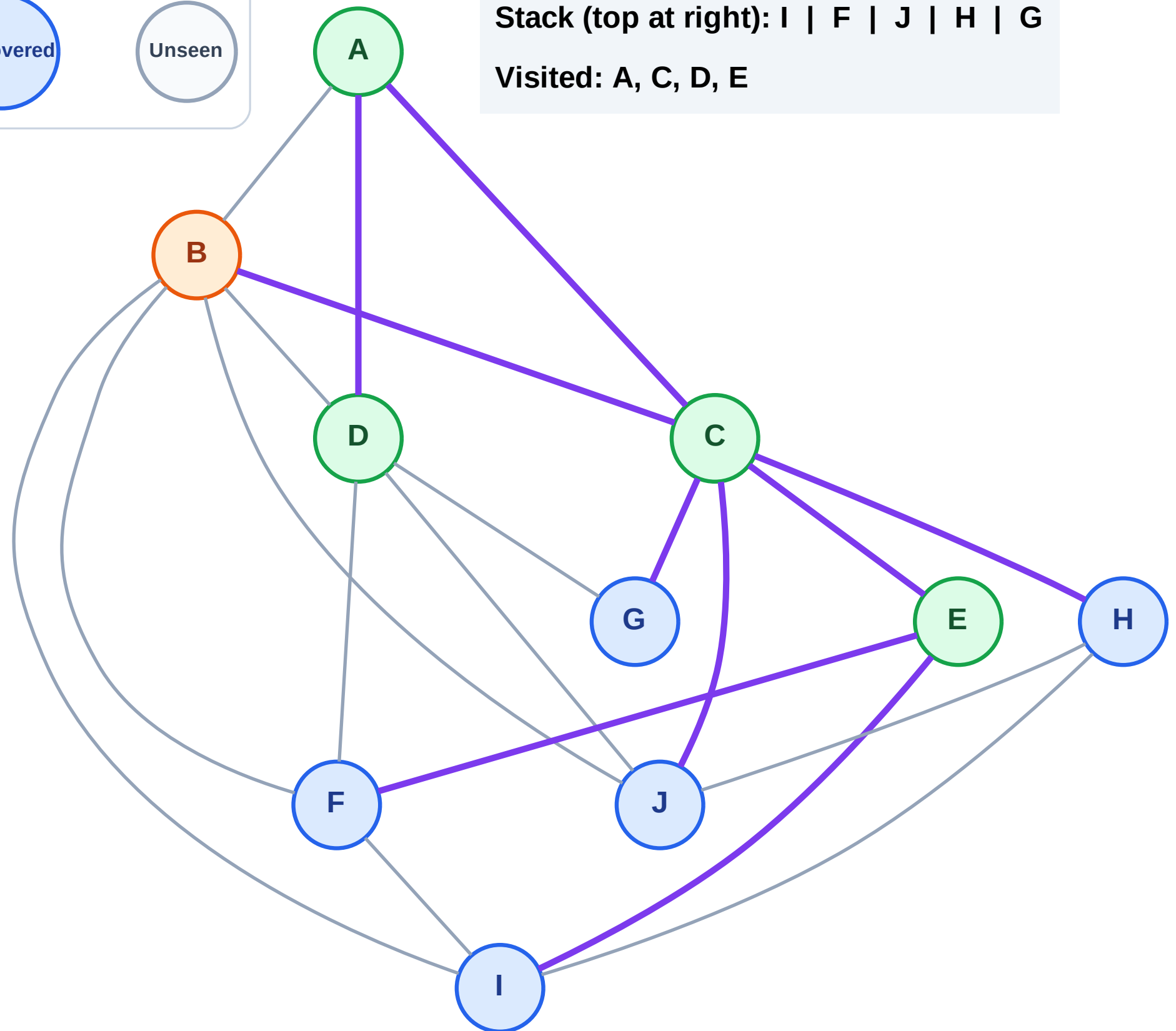
Step 10: Process node B

Step 10: Process node B



Stack (top at right): I | F | J | H | G

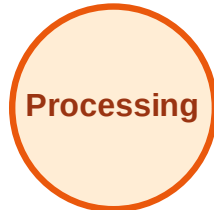
Visited: A, C, D, E



DFS Traversal

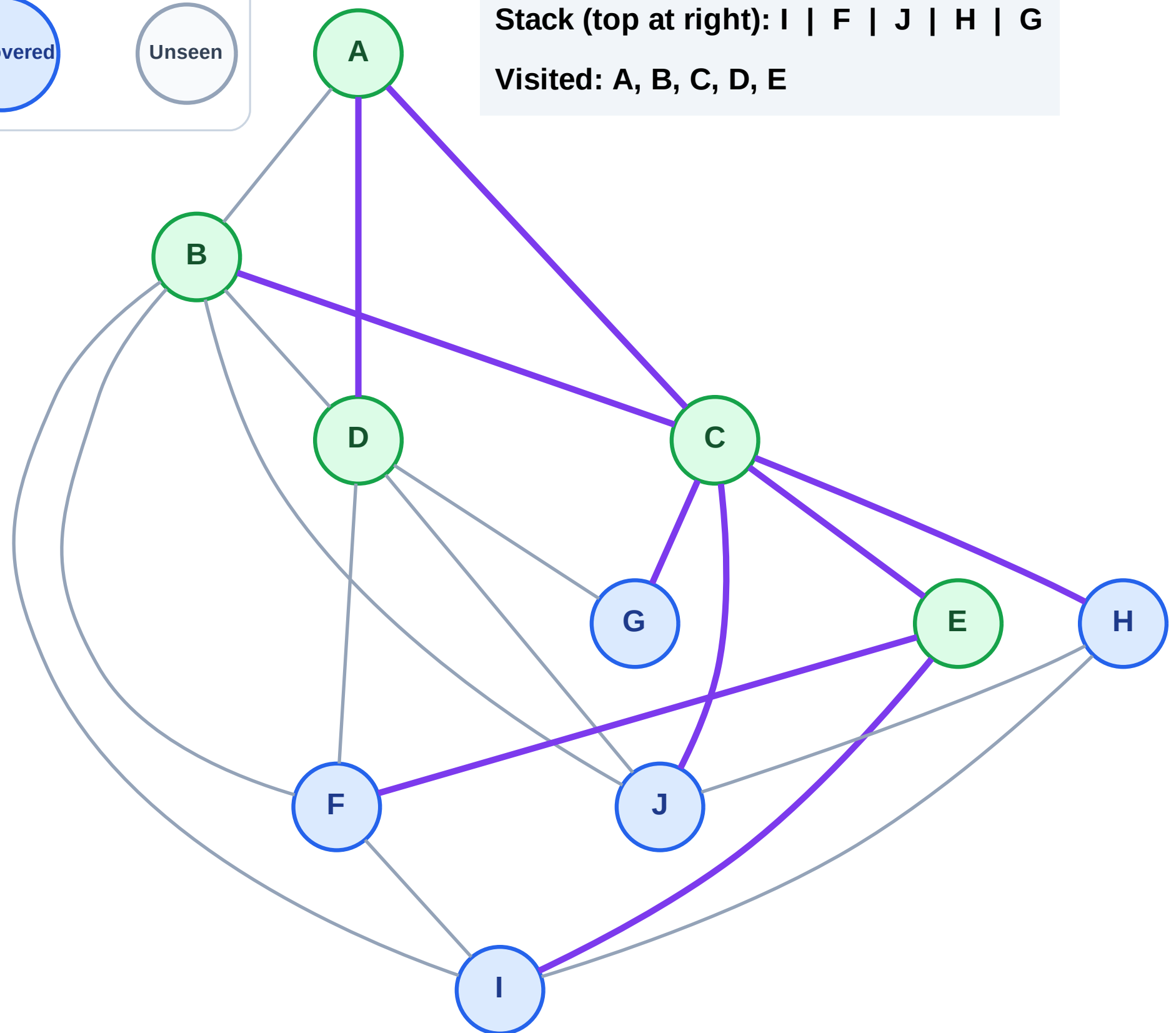
Step 11: No new neighbors from B

Legend



Stack (top at right): I | F | J | H | G

Visited: A, B, C, D, E



DFS Traversal

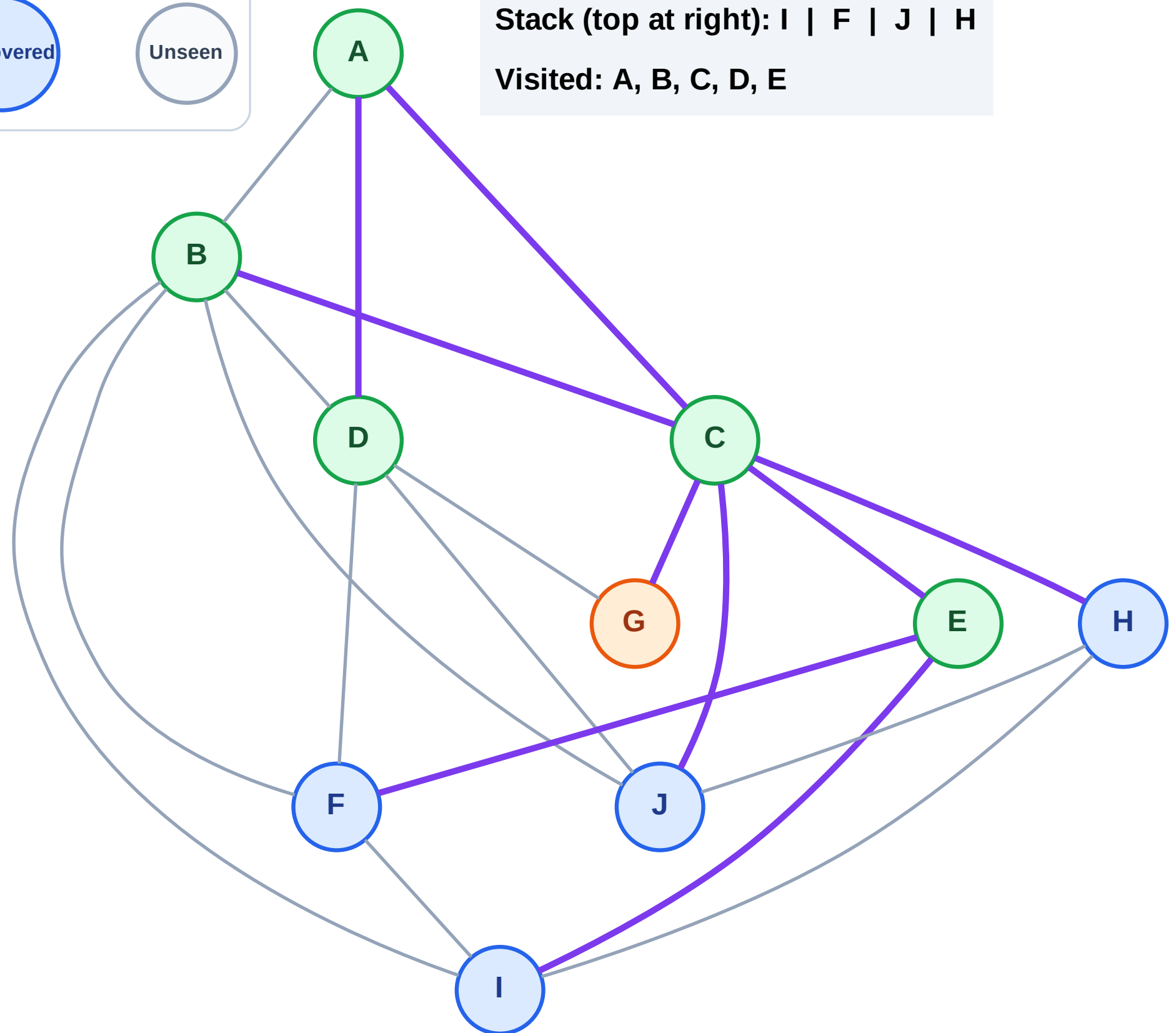
Step 12: Process node G

Legend



Stack (top at right): I | F | J | H

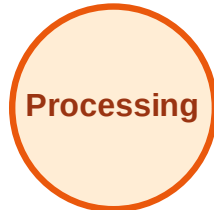
Visited: A, B, C, D, E



DFS Traversal

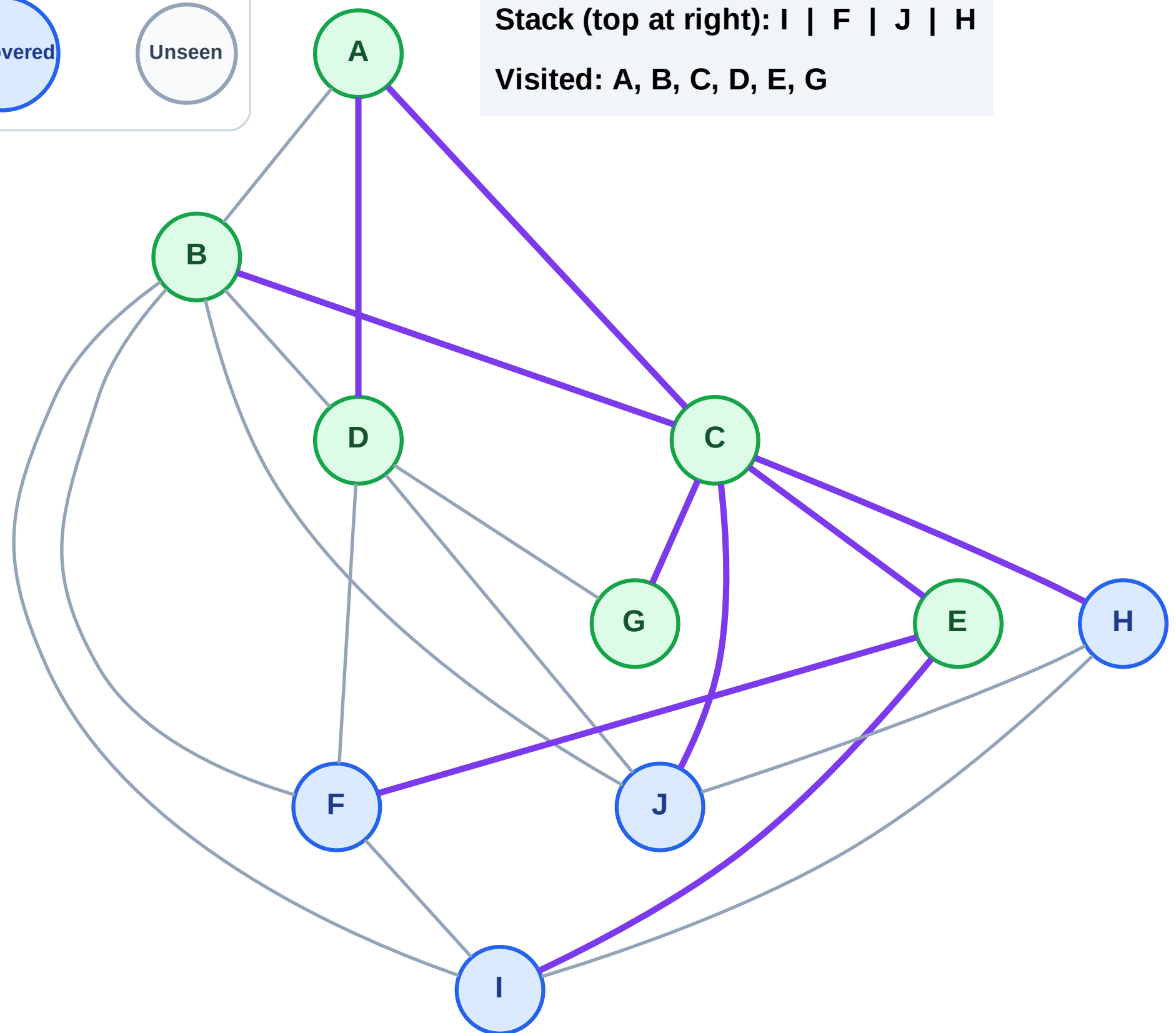
Step 13: No new neighbors from G

Legend



Stack (top at right): I | F | J | H

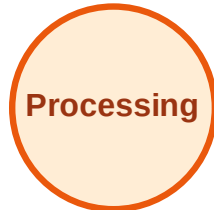
Visited: A, B, C, D, E, G



DFS Traversal

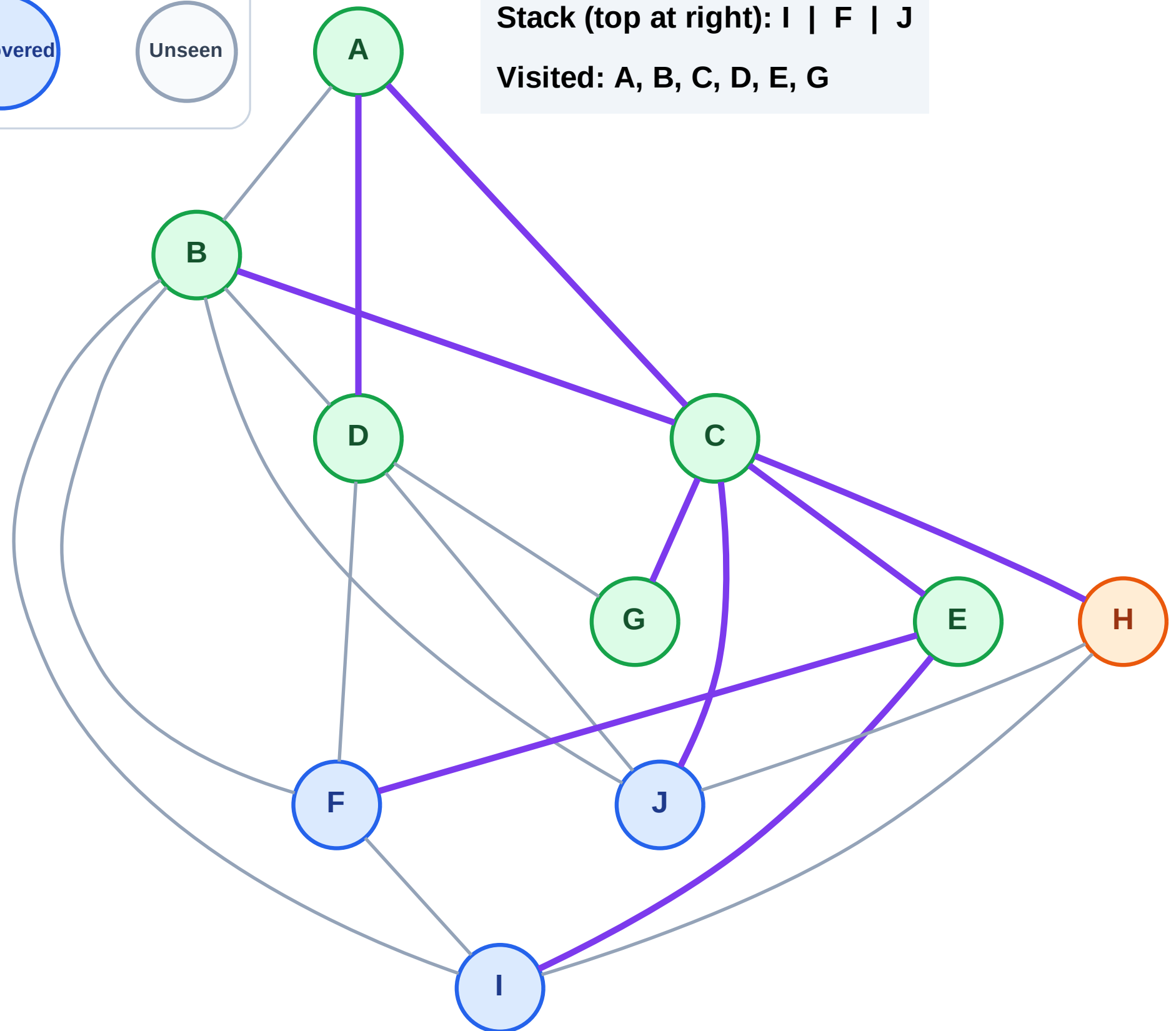
Step 14: Process node H

Legend



Stack (top at right): I | F | J

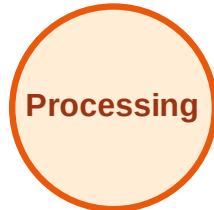
Visited: A, B, C, D, E, G



DFS Traversal

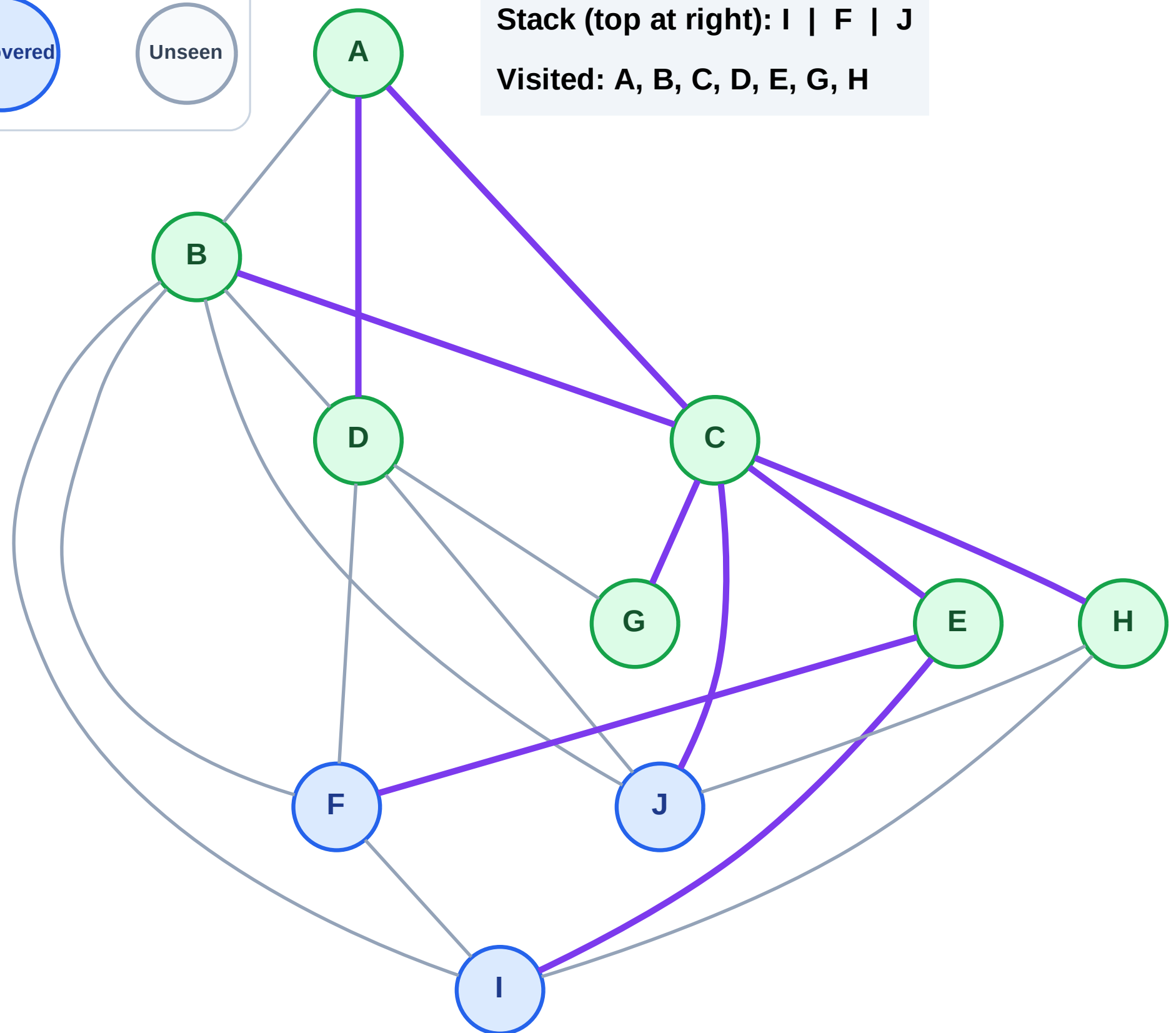
Step 15: No new neighbors from H

Legend



Stack (top at right): I | F | J

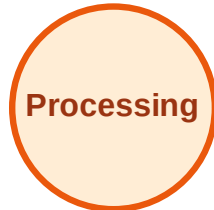
Visited: A, B, C, D, E, G, H



DFS Traversal

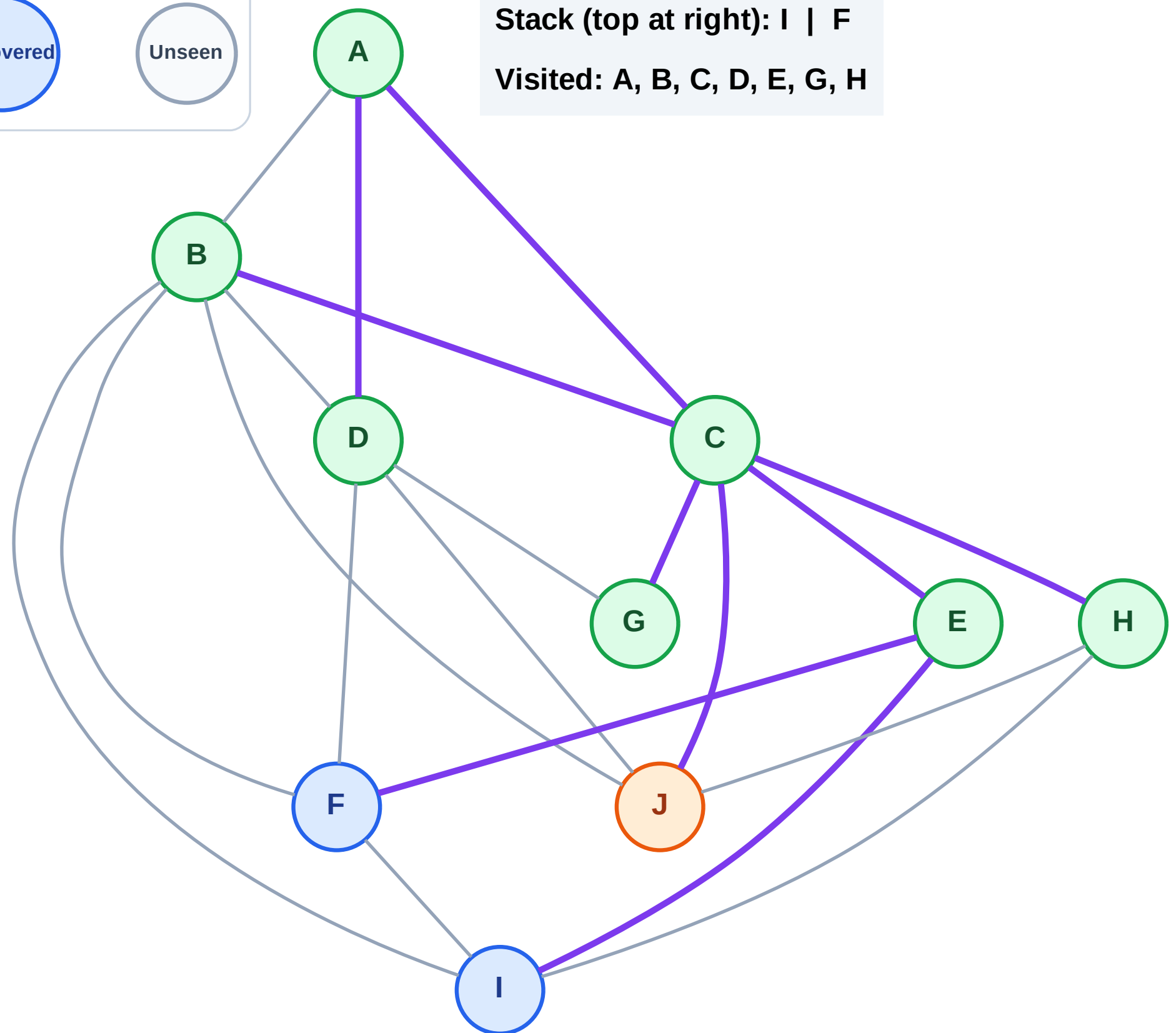
Step 16: Process node J

Legend



Stack (top at right): I | F

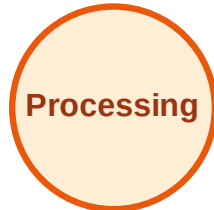
Visited: A, B, C, D, E, G, H



DFS Traversal

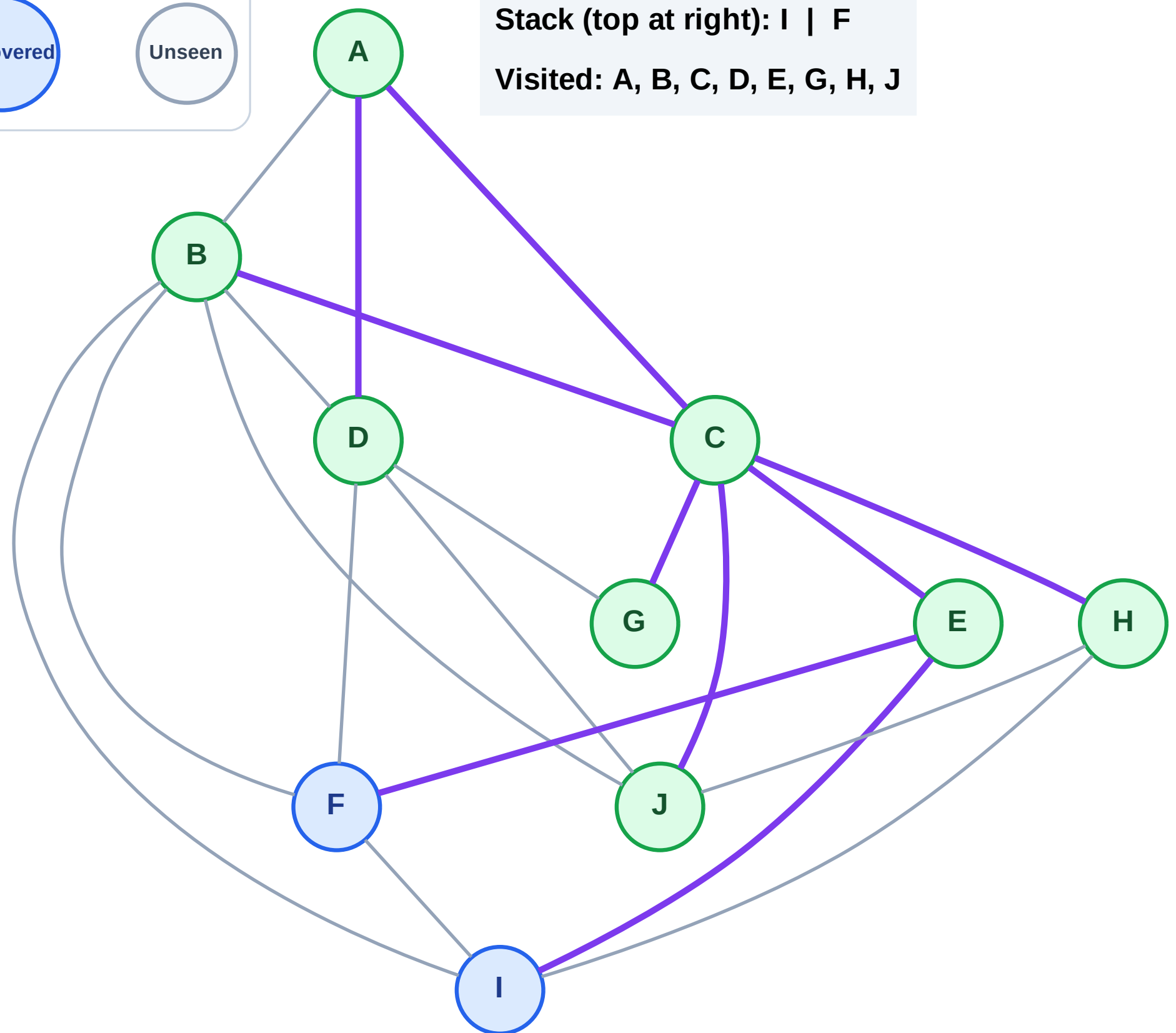
Step 17: No new neighbors from J

Legend



Stack (top at right): I | F

Visited: A, B, C, D, E, G, H, J



DFS Traversal

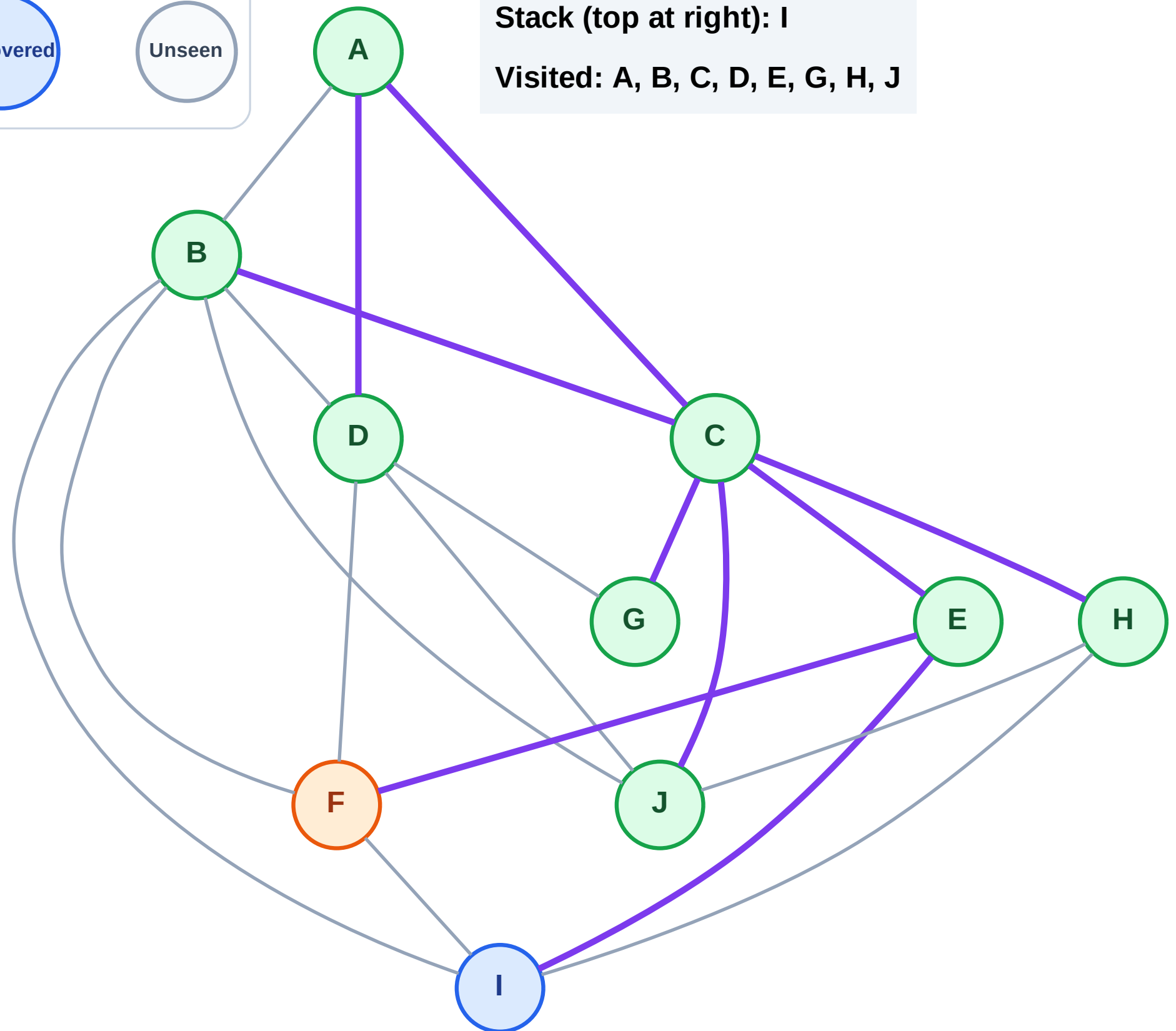
Step 18: Process node F

Legend



Stack (top at right): I

Visited: A, B, C, D, E, G, H, J

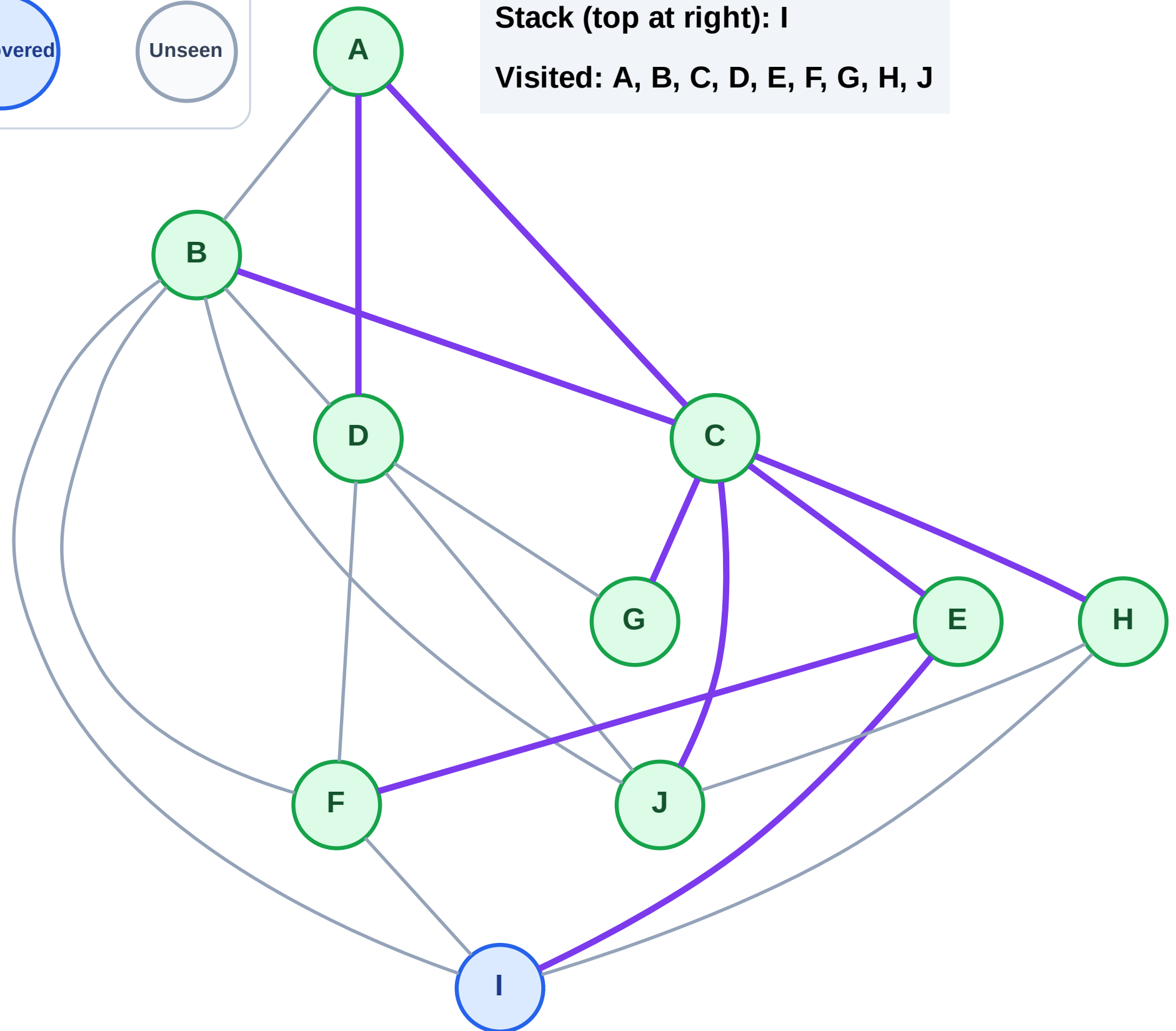


Step 19: No new neighbors from F

Legend

Unseen

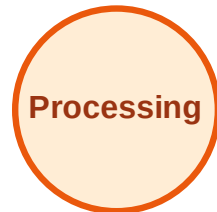
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

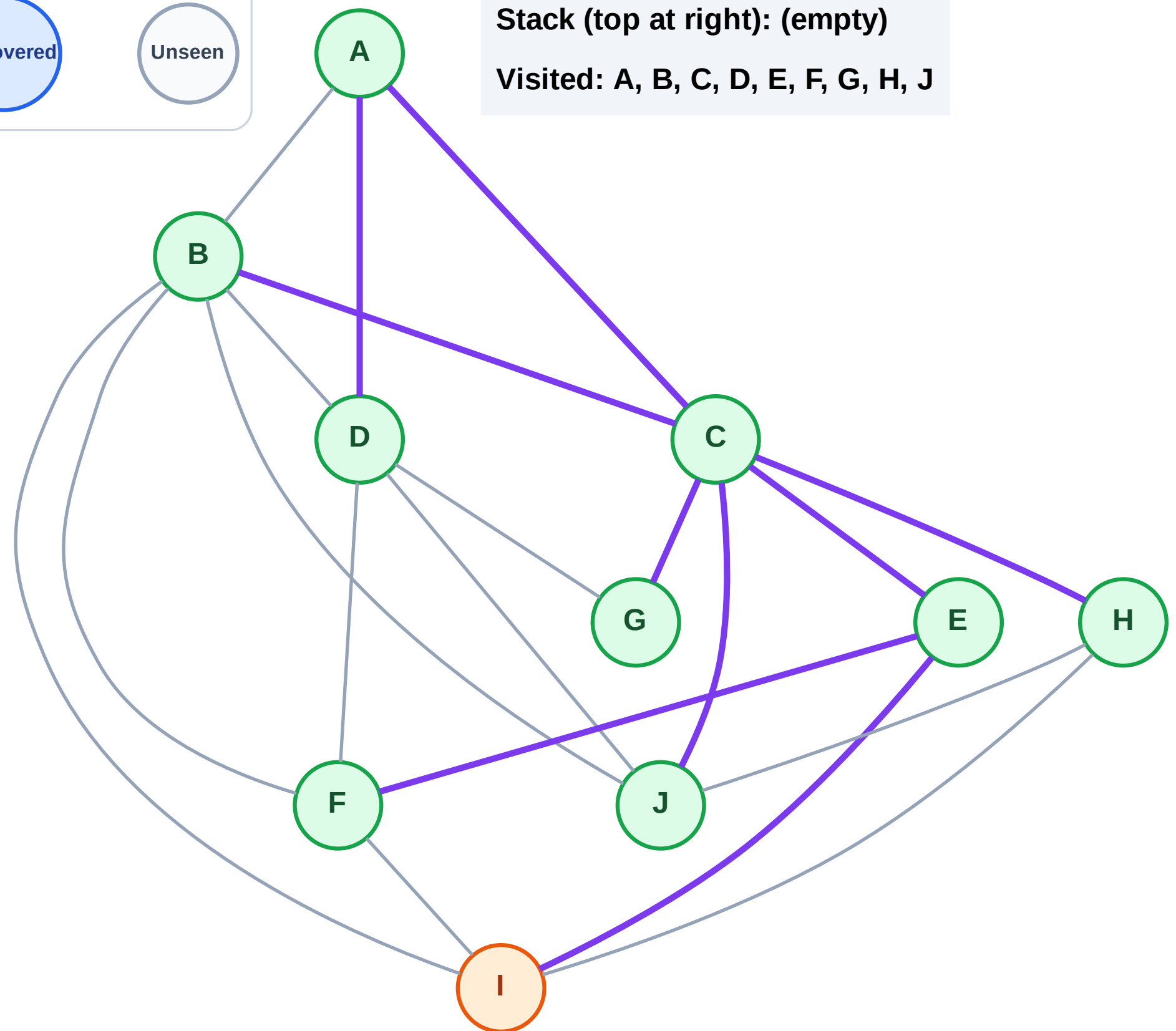
Step 20: Process node I

Legend



Stack (top at right): (empty)

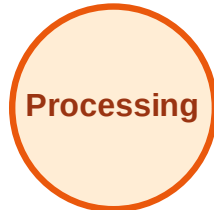
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

